

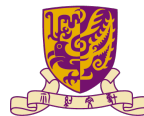
Network Programming

Collective Communication Algorithm

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SCHOOL OF
DATA SCIENCE
數據科學學院

Recap: Collective communication primitives

- Broadcast
- Reduce(-to-one)
- Scatter
- Gather
- Allgather
- Reduce-scatter
- Allreduce

Model

- a node can send directly to any other node (maybe not true)
- a node can simultaneously receive and send
- cost of communication
 - sending a message of length n between any two

$$\alpha + n \beta$$

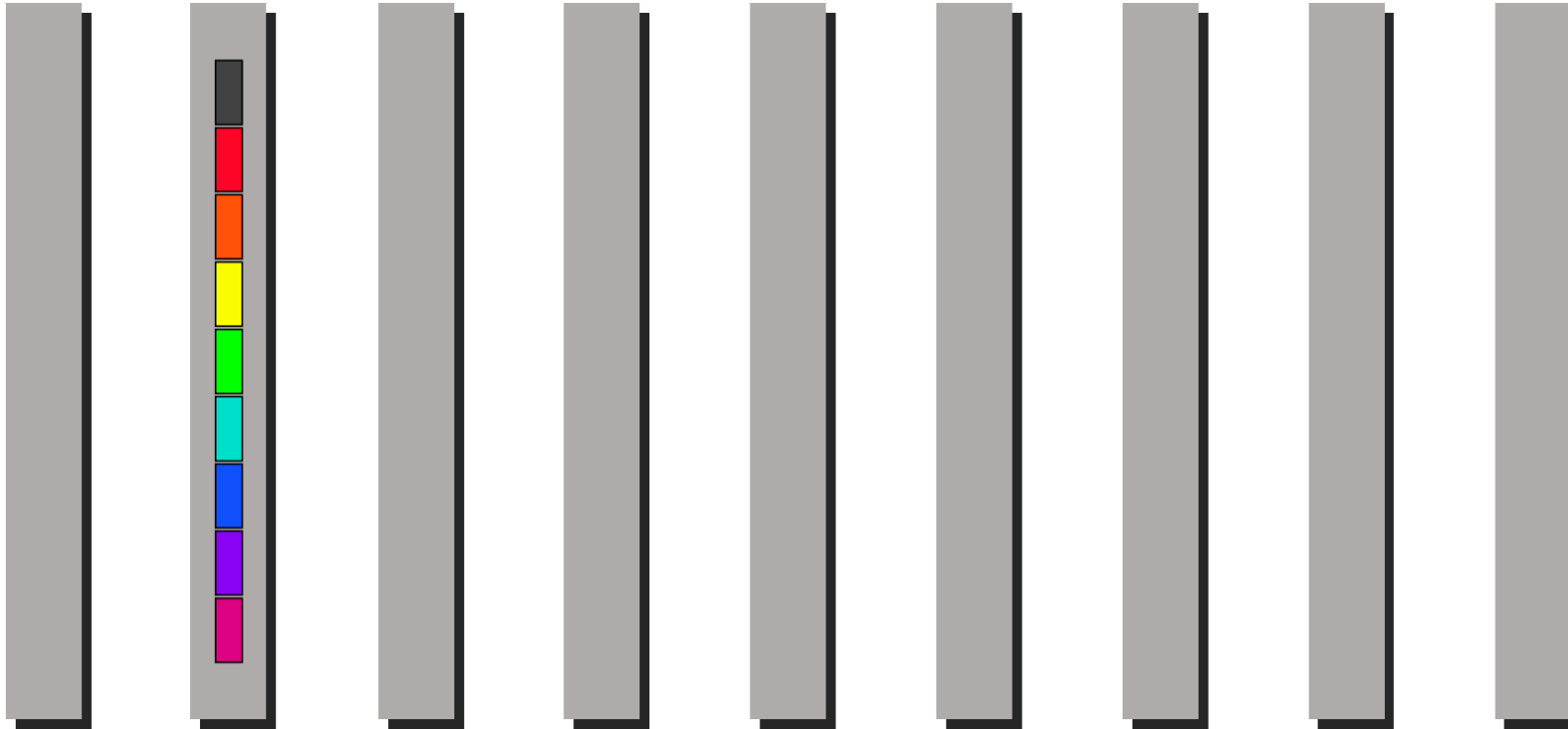
- if a message encounters a link that simultaneously accommodates M messages, the cost becomes

$$\alpha + Mn \beta$$

Two Family of Mainstream Algorithms

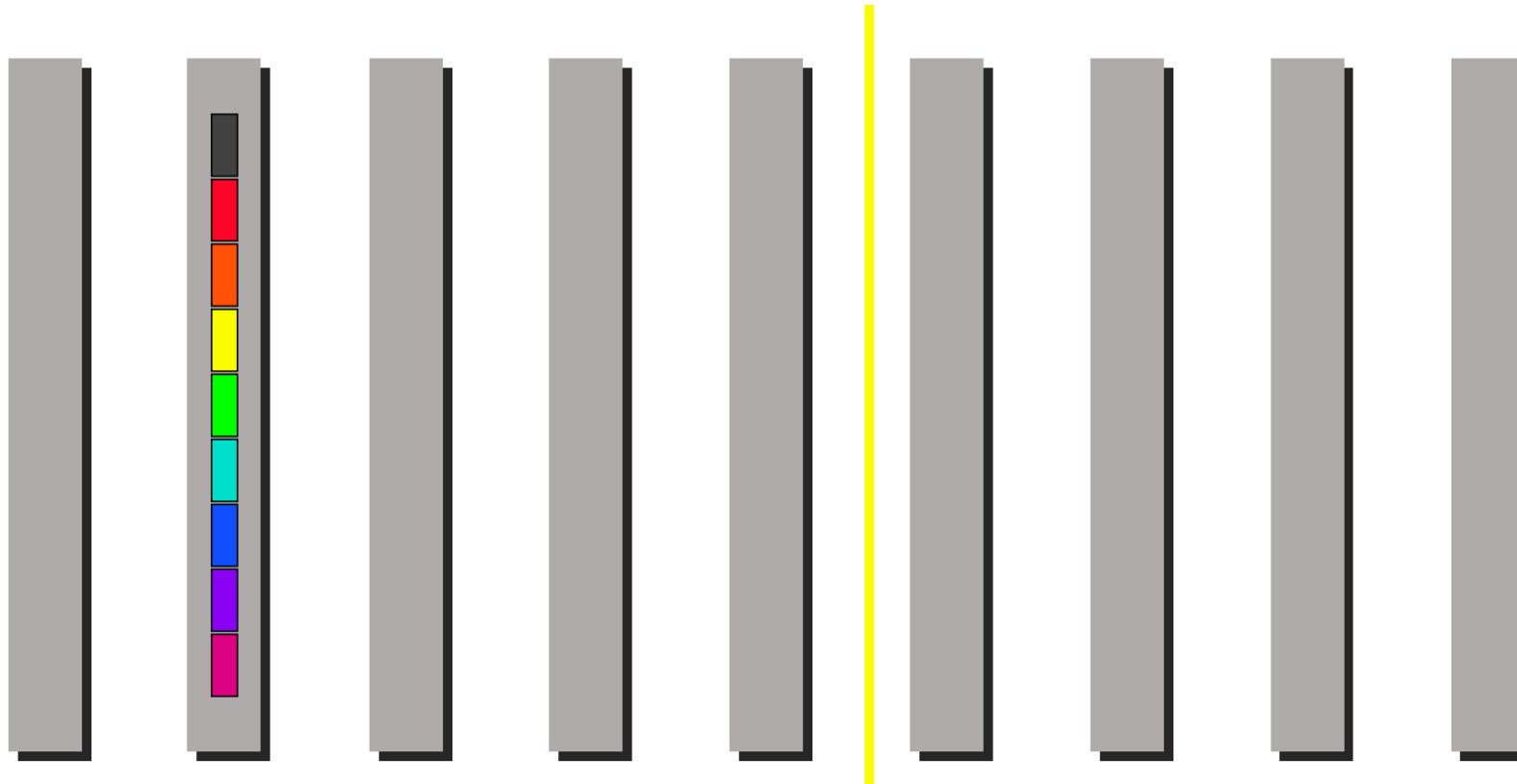
- Small message: Minimum Spanning Tree algorithm
 - Emphasize **low latency**
- Large Message: Ring algorithm
 - Emphasize **bandwidth utilization**
- There are 50+ different algorithms developed in the past 50 years

General principles



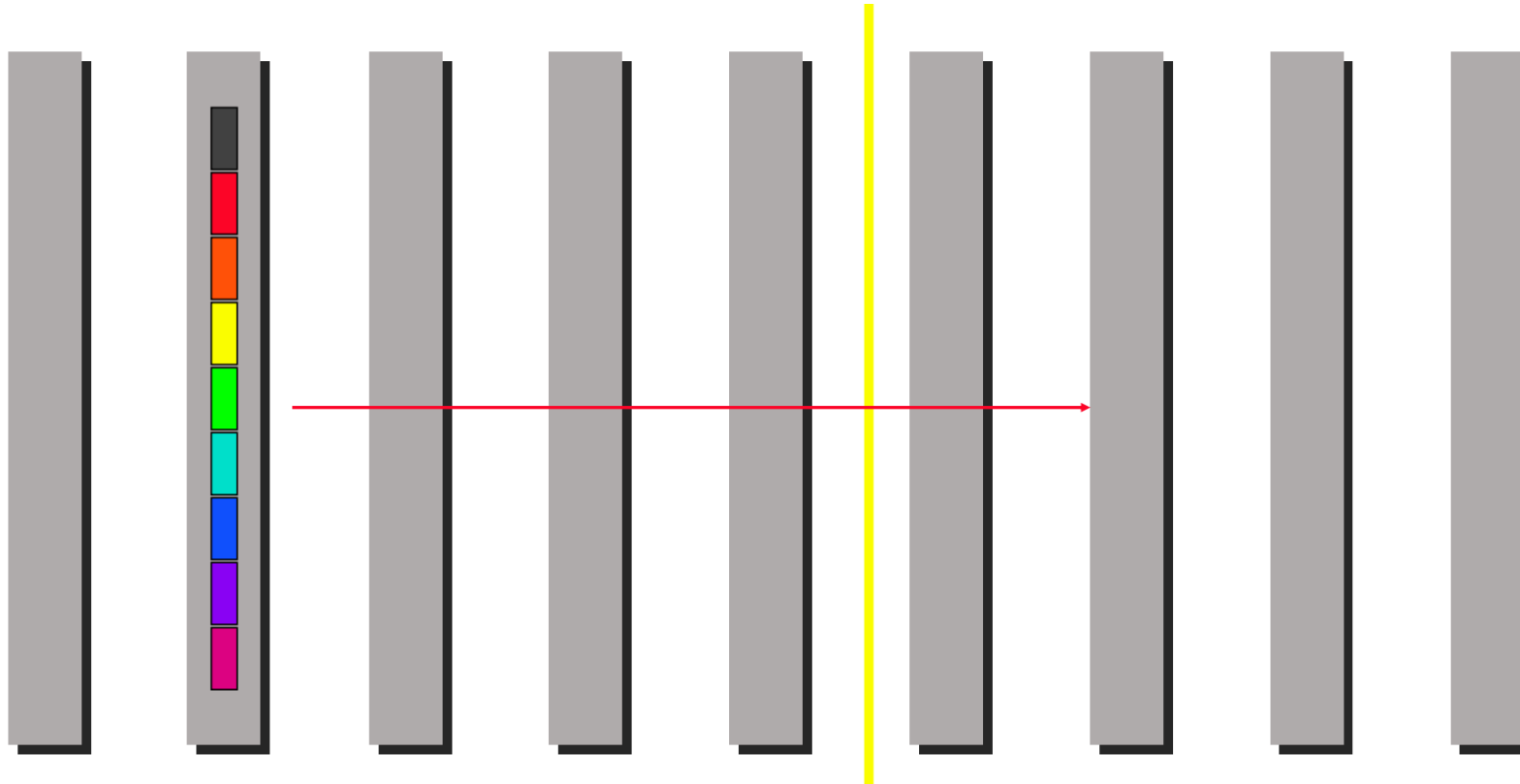
- Message starts on one worker/process

General principles



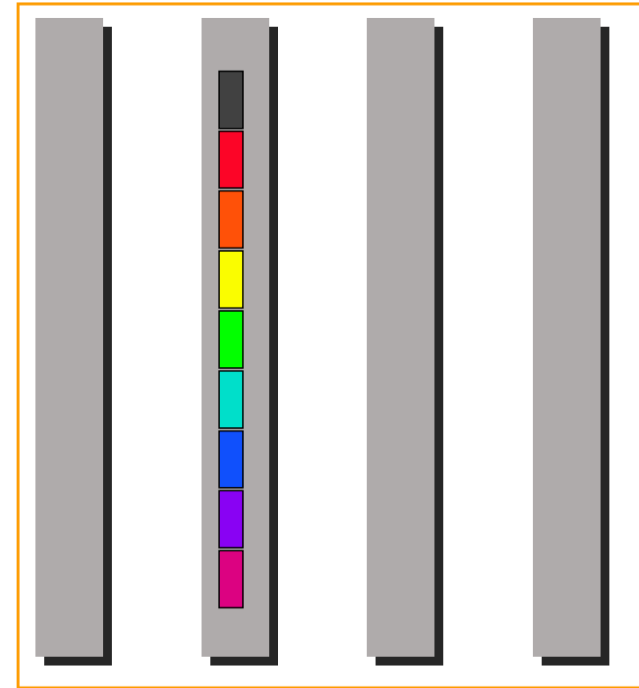
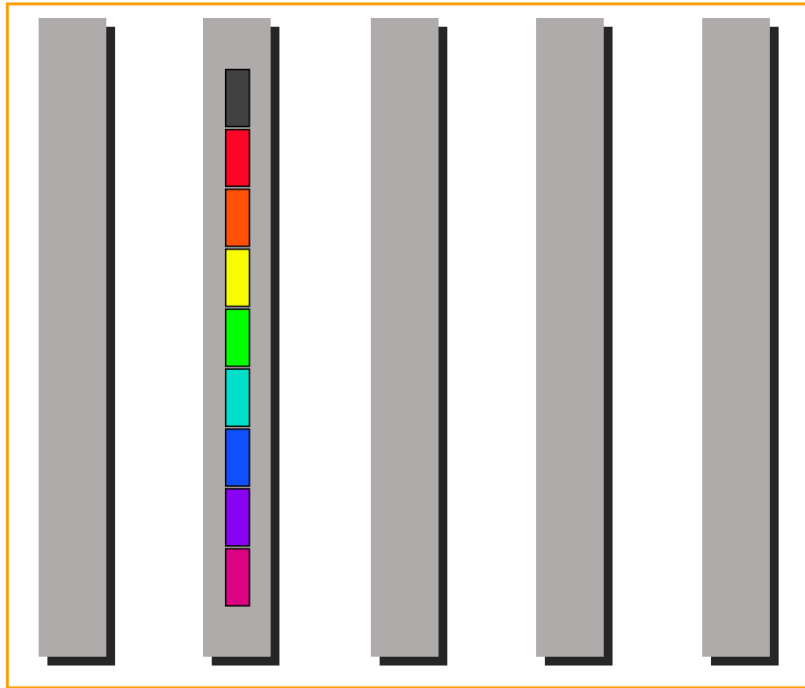
- Divide logical linear array in half

General principles



- Send message to the half of the network that does not contain the current node (root) that holds the message

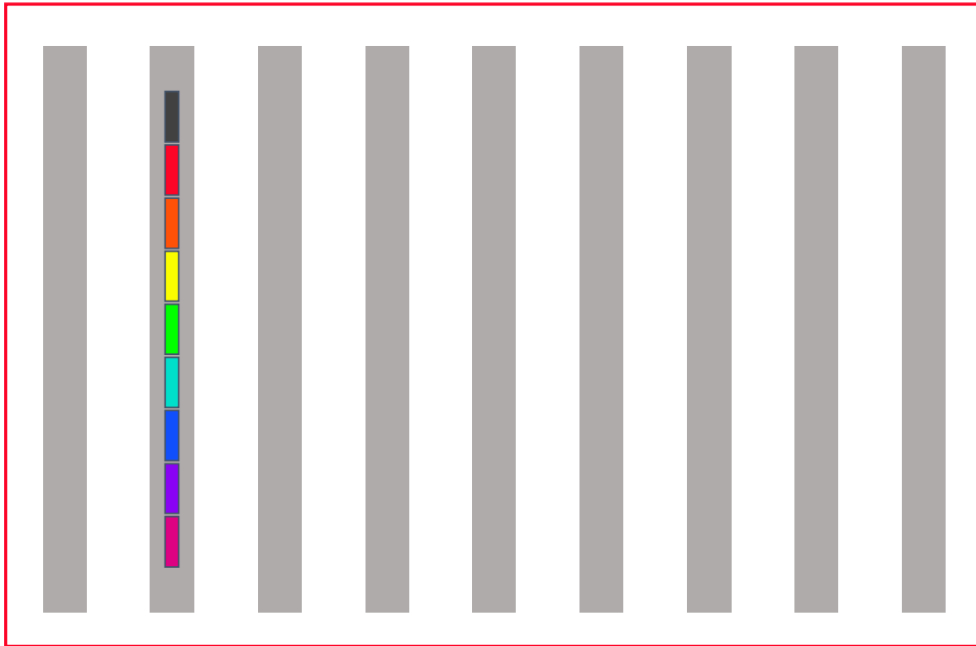
General principles



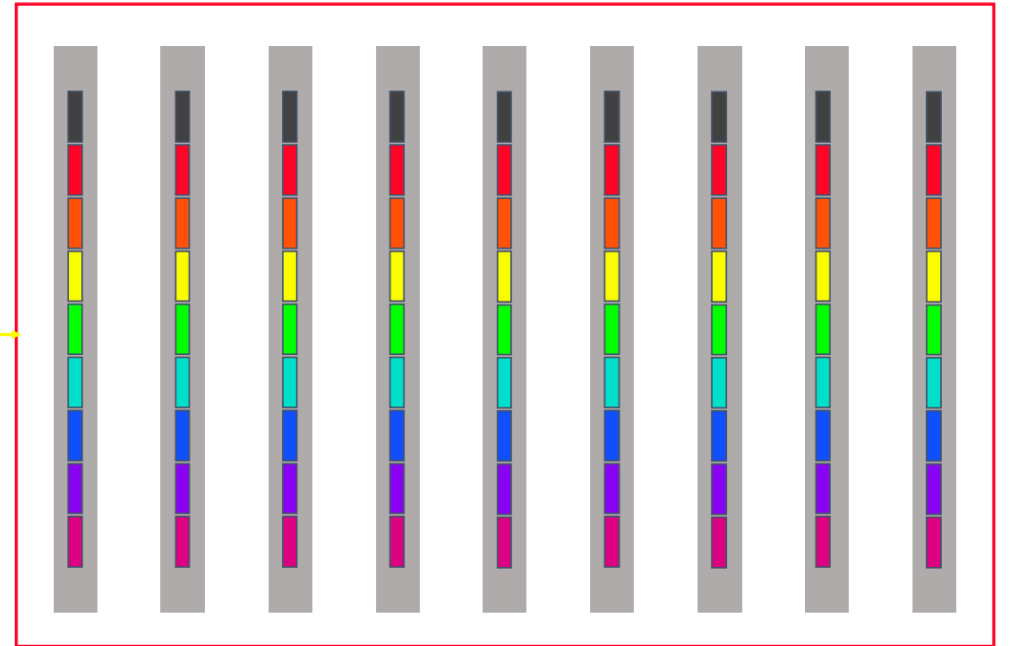
- Continue recursively in each of the two halves

Broadcast

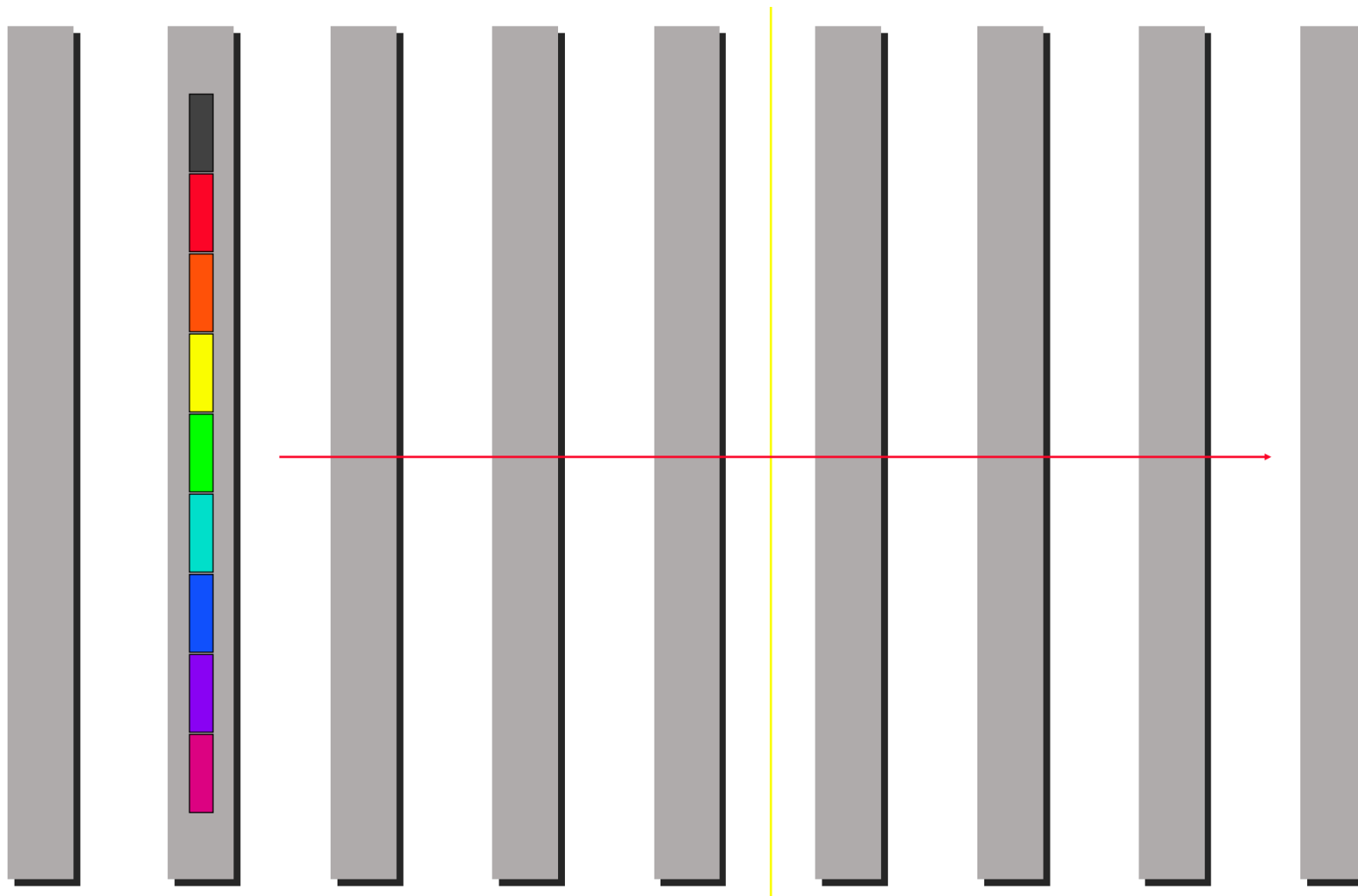
Before



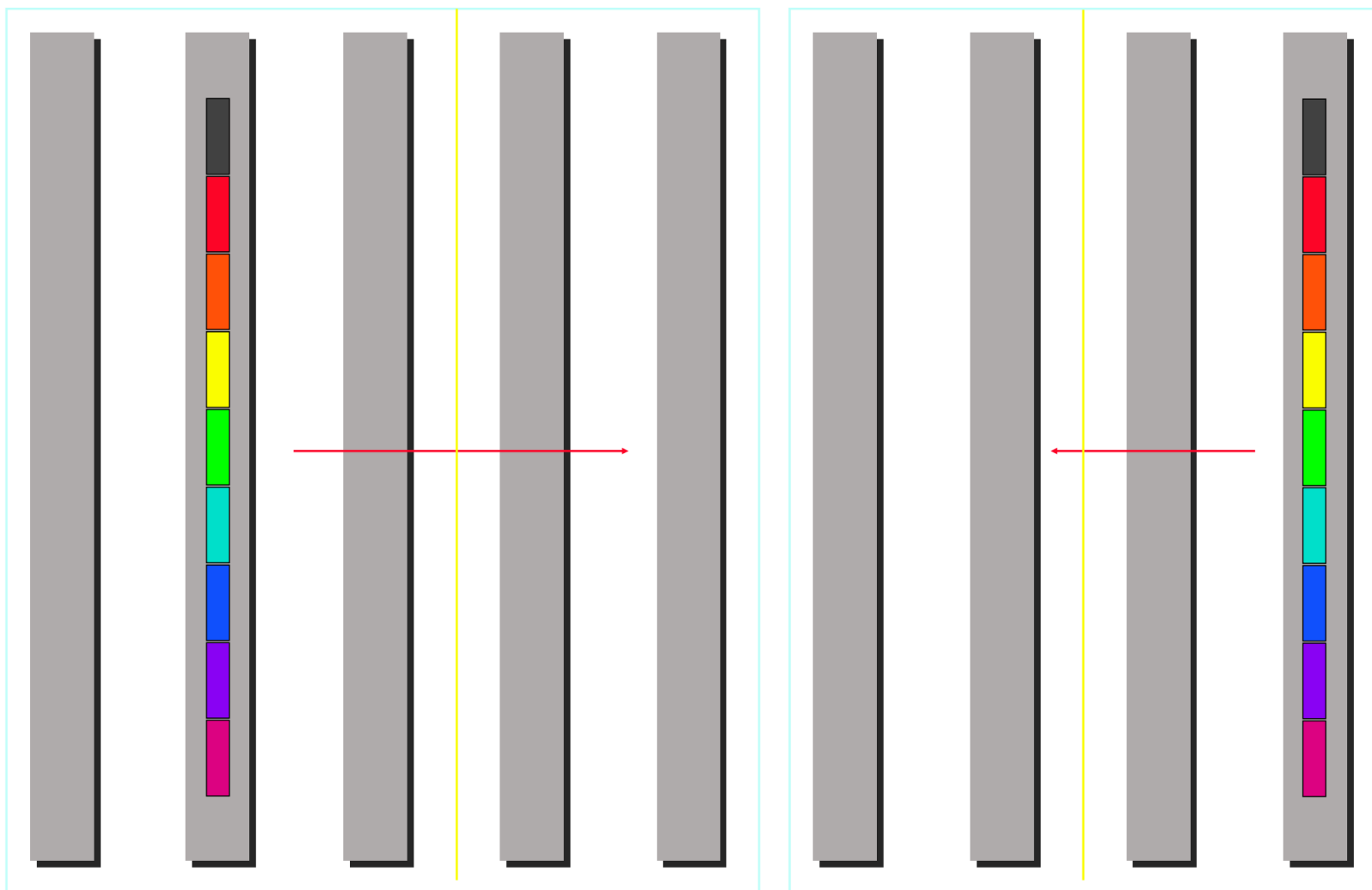
After



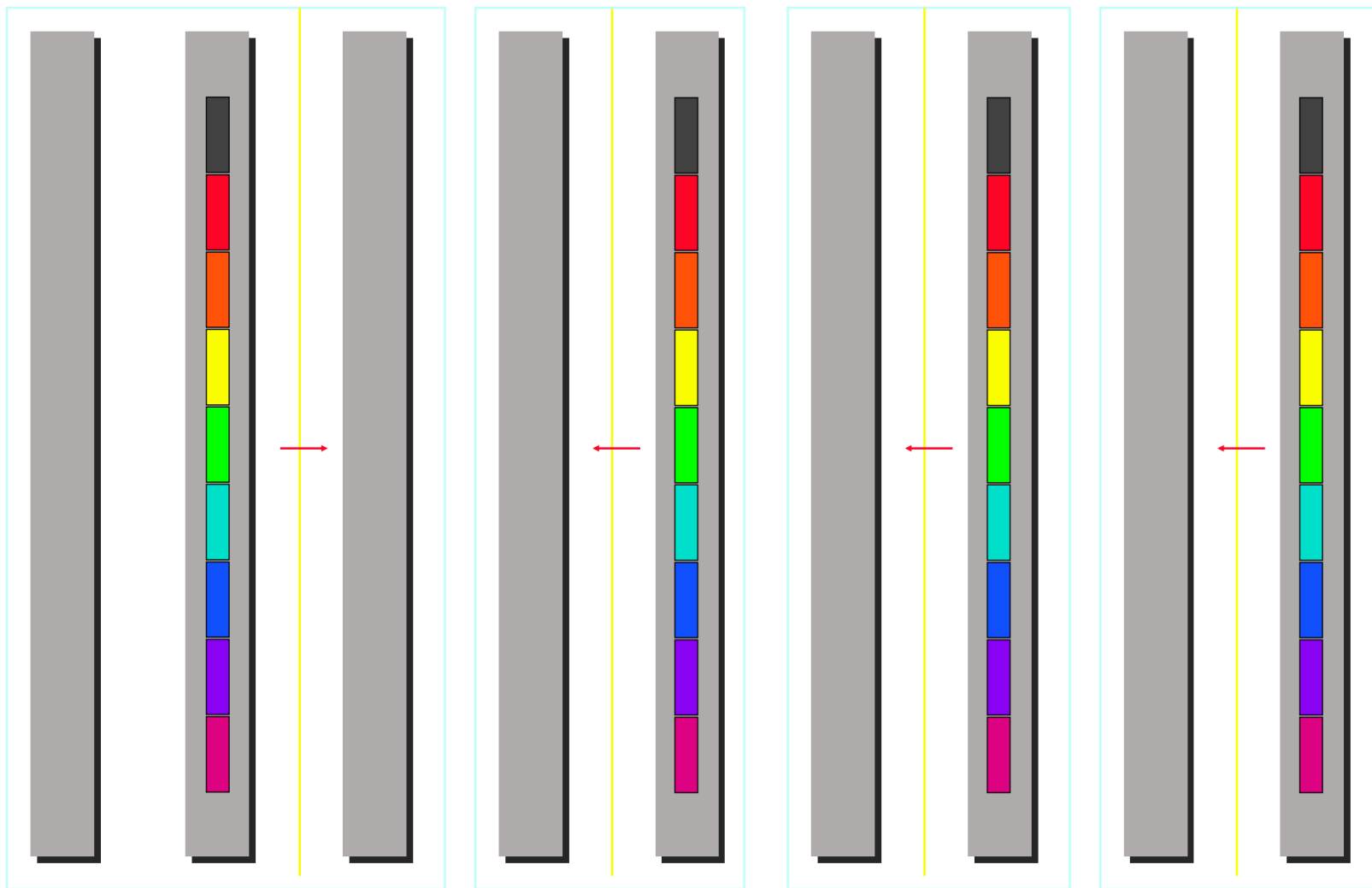
Broadcast(1)



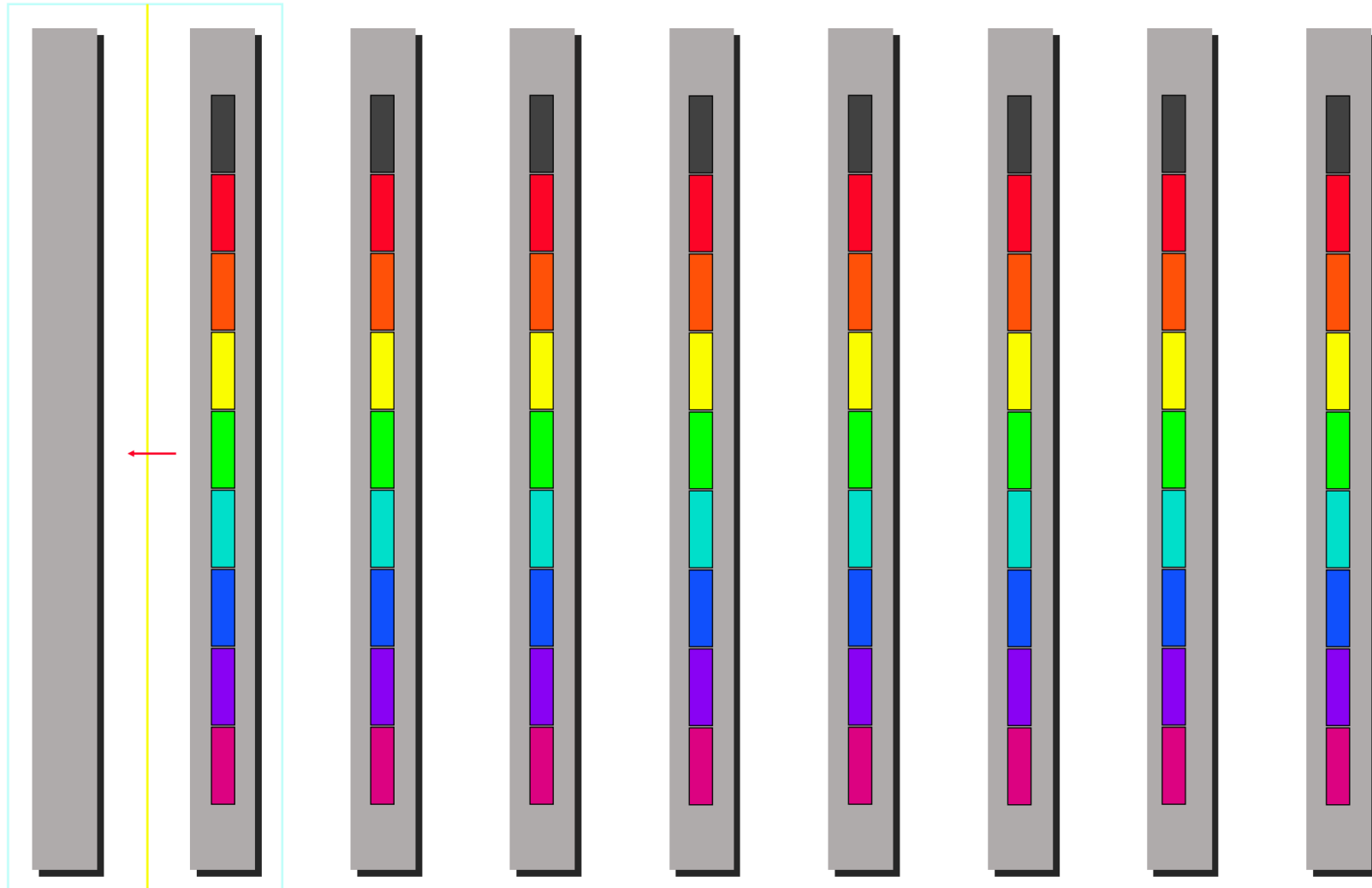
Broadcast(2)



Broadcast(3)



Broadcast(4)

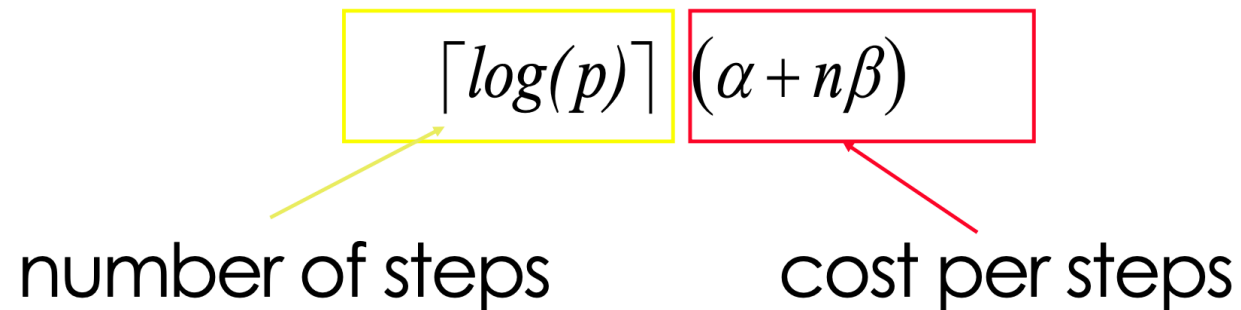


Cost of minimum spanning tree broadcast

- Latency of communication startup (α)
- Transmission delay in proportion to data size ($n\beta$)
- Total cost:

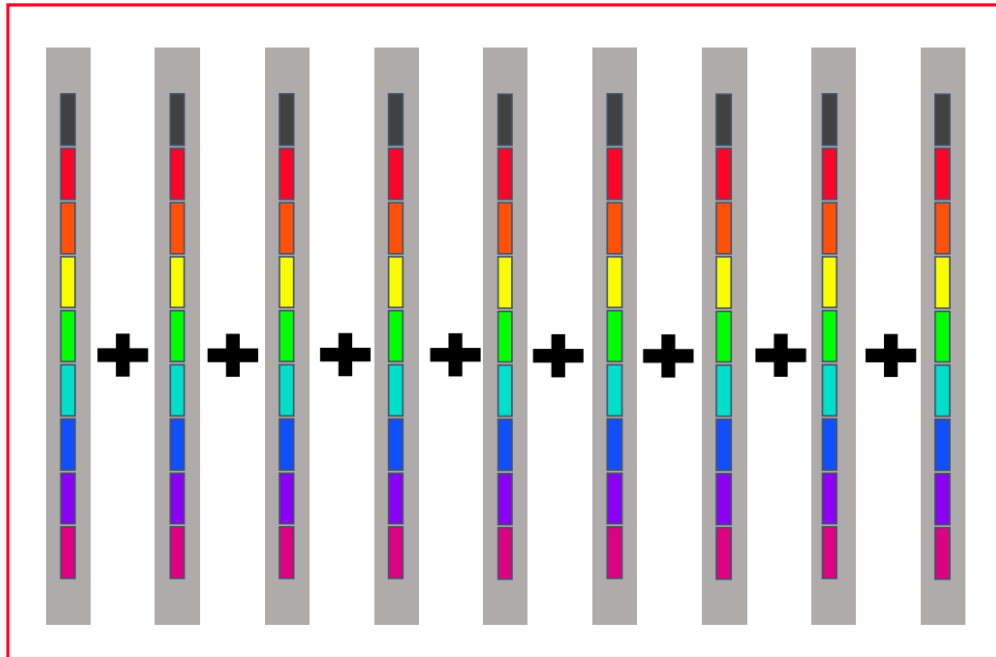
$$\boxed{\lceil \log(p) \rceil} \boxed{(\alpha + n\beta)}$$

number of steps cost per steps

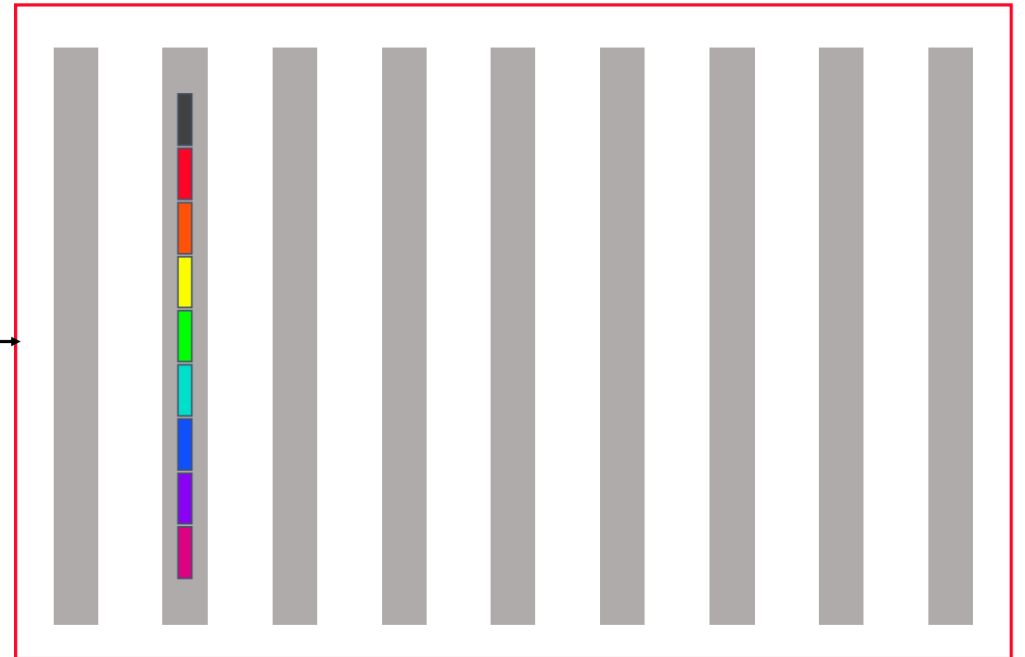


Reduce

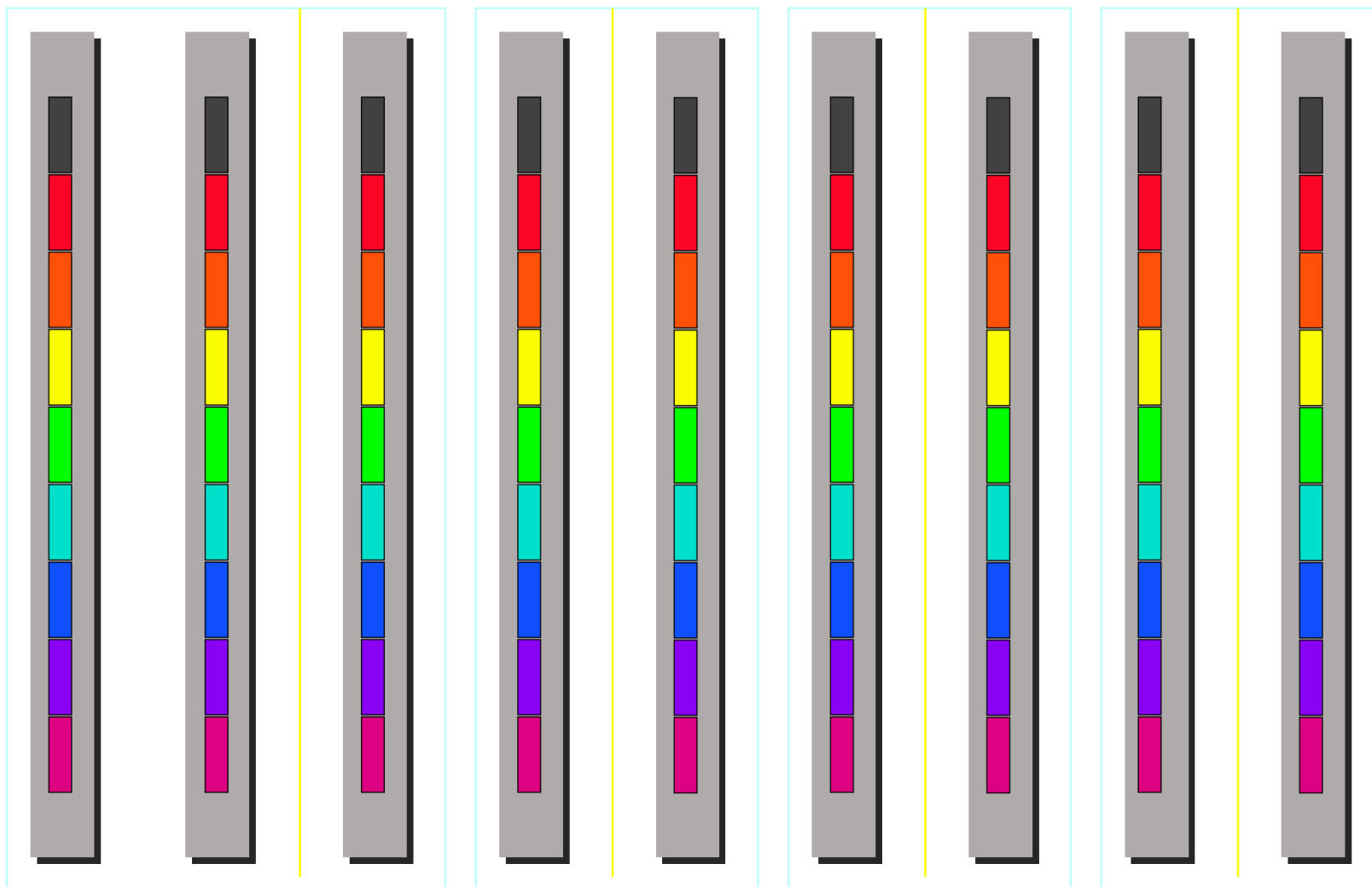
Before



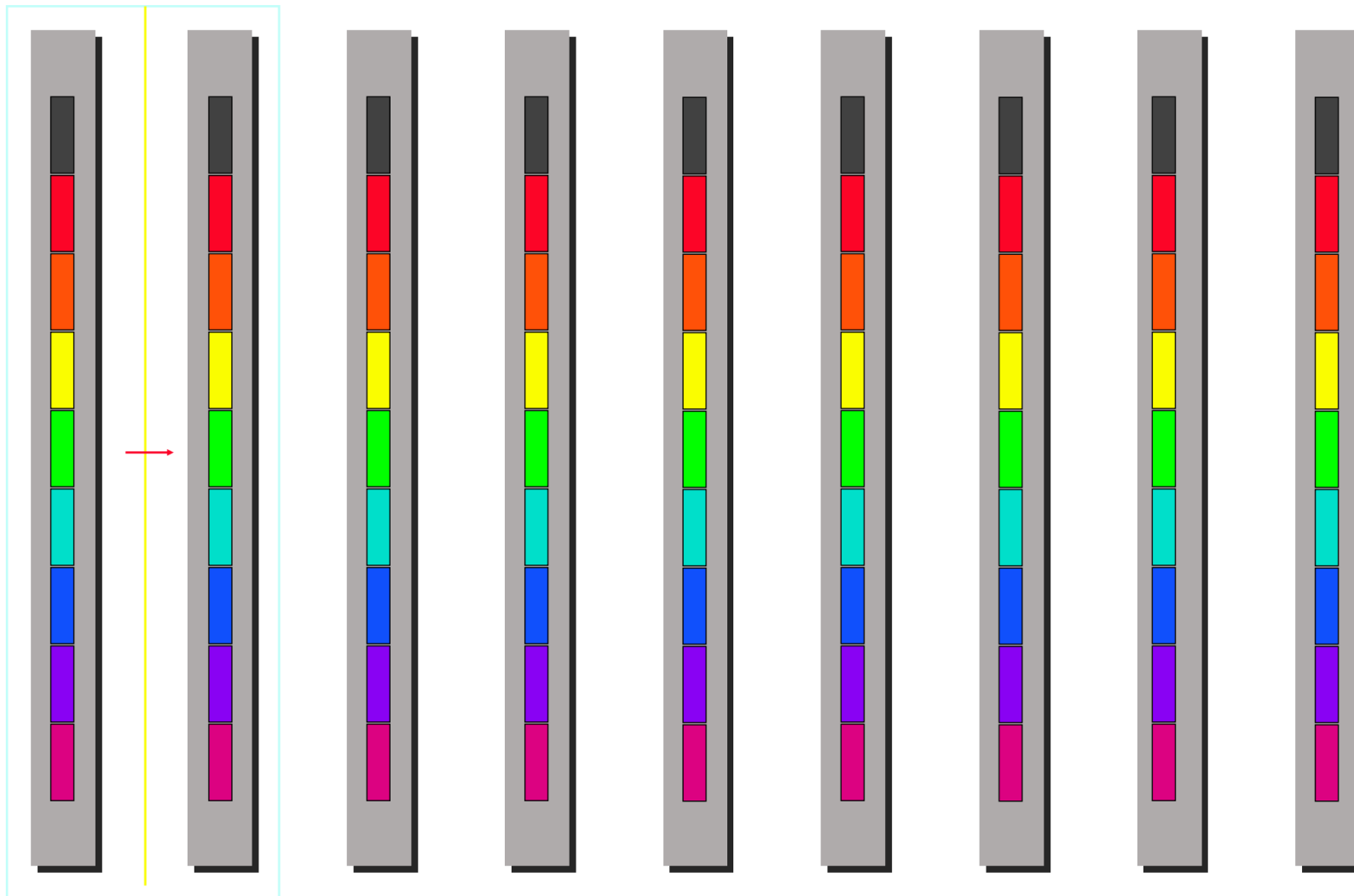
After



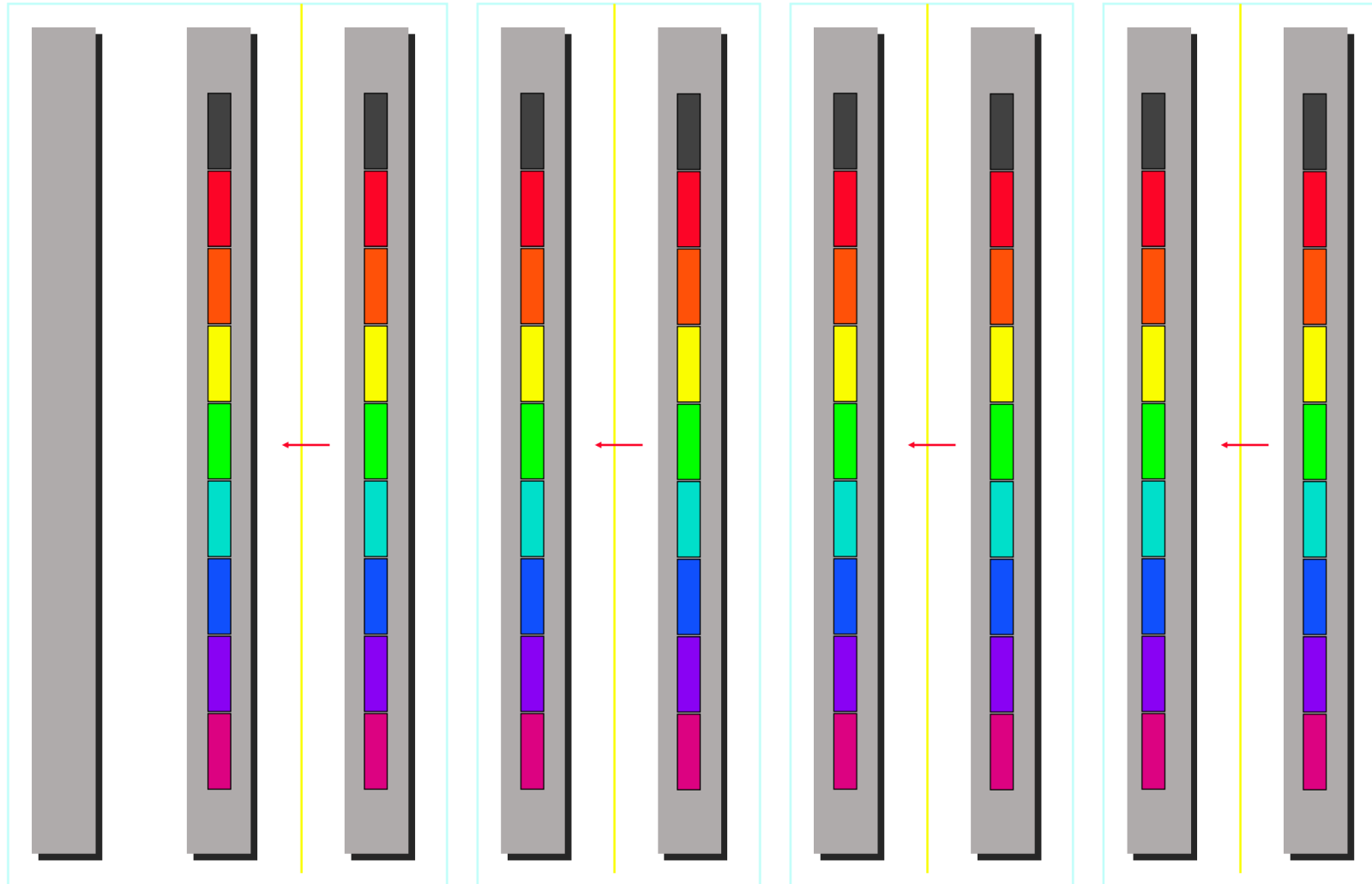
Reduce(1)



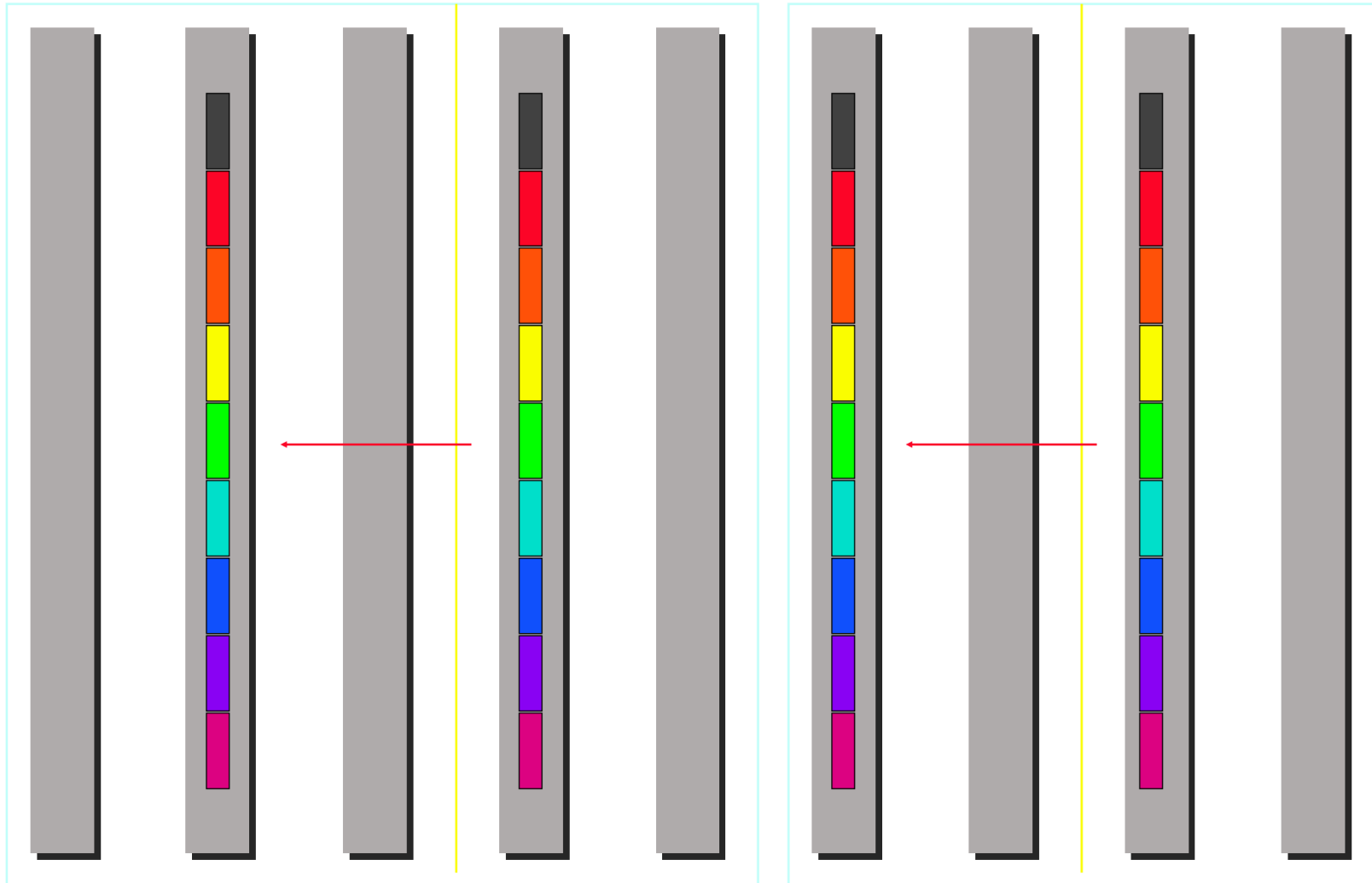
Reduce(2)



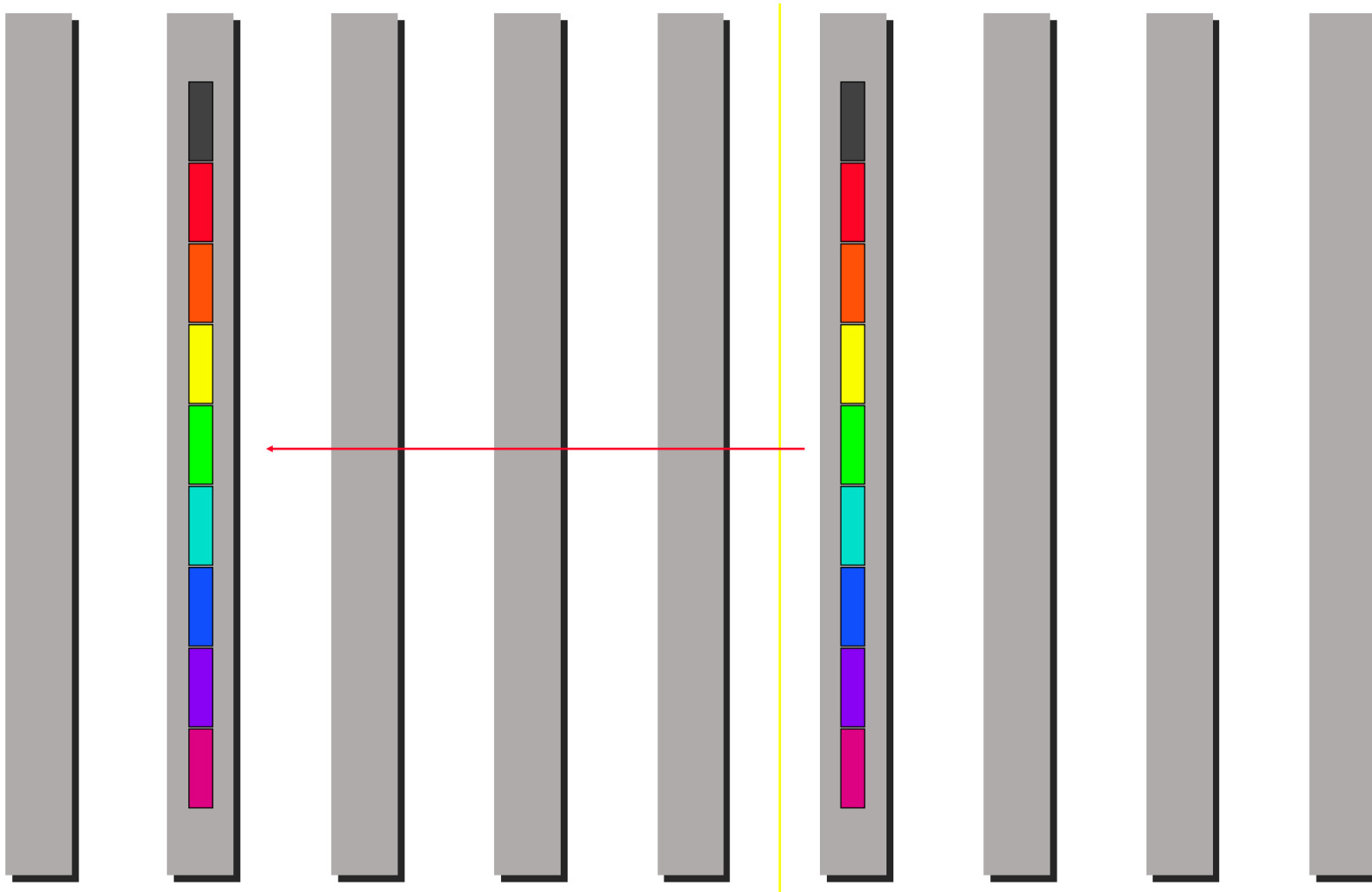
Reduce(3)



Reduce(4)



Reduce(5)



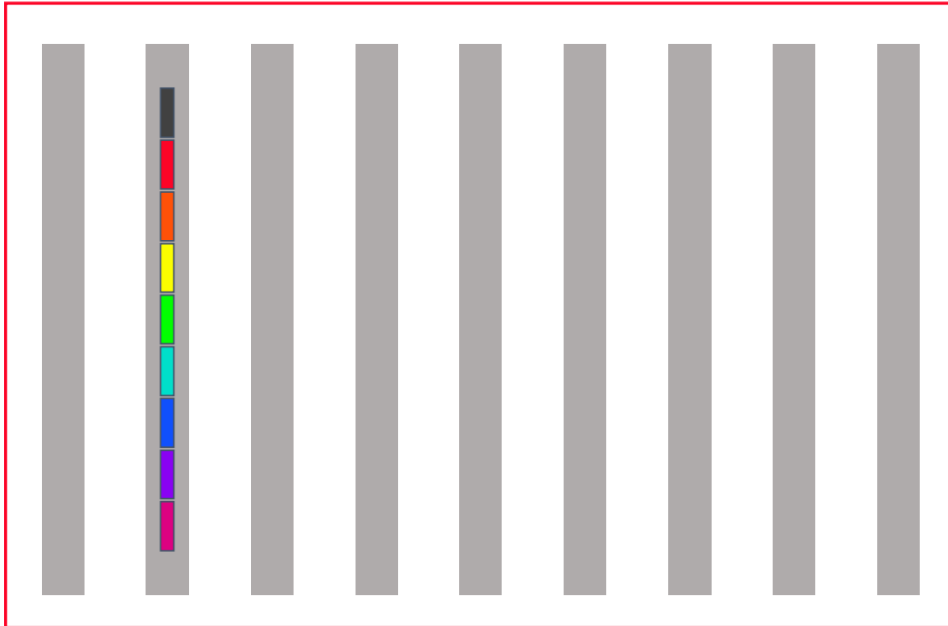
Cost of minimum spanning tree reduce

$$\lceil \log(p) \rceil (\alpha + n\beta + n\gamma)$$

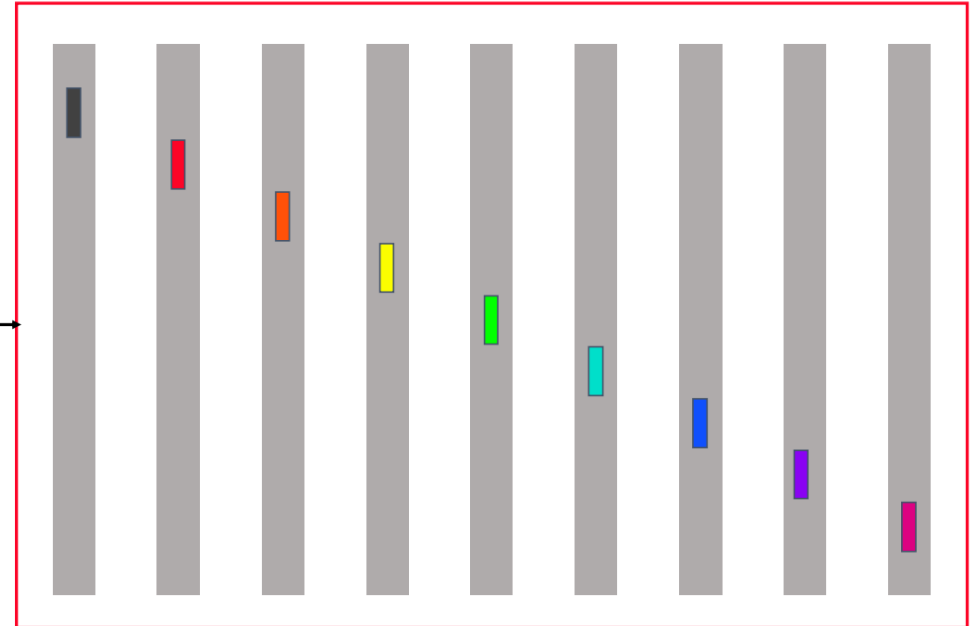
Compute overhead

Scatter

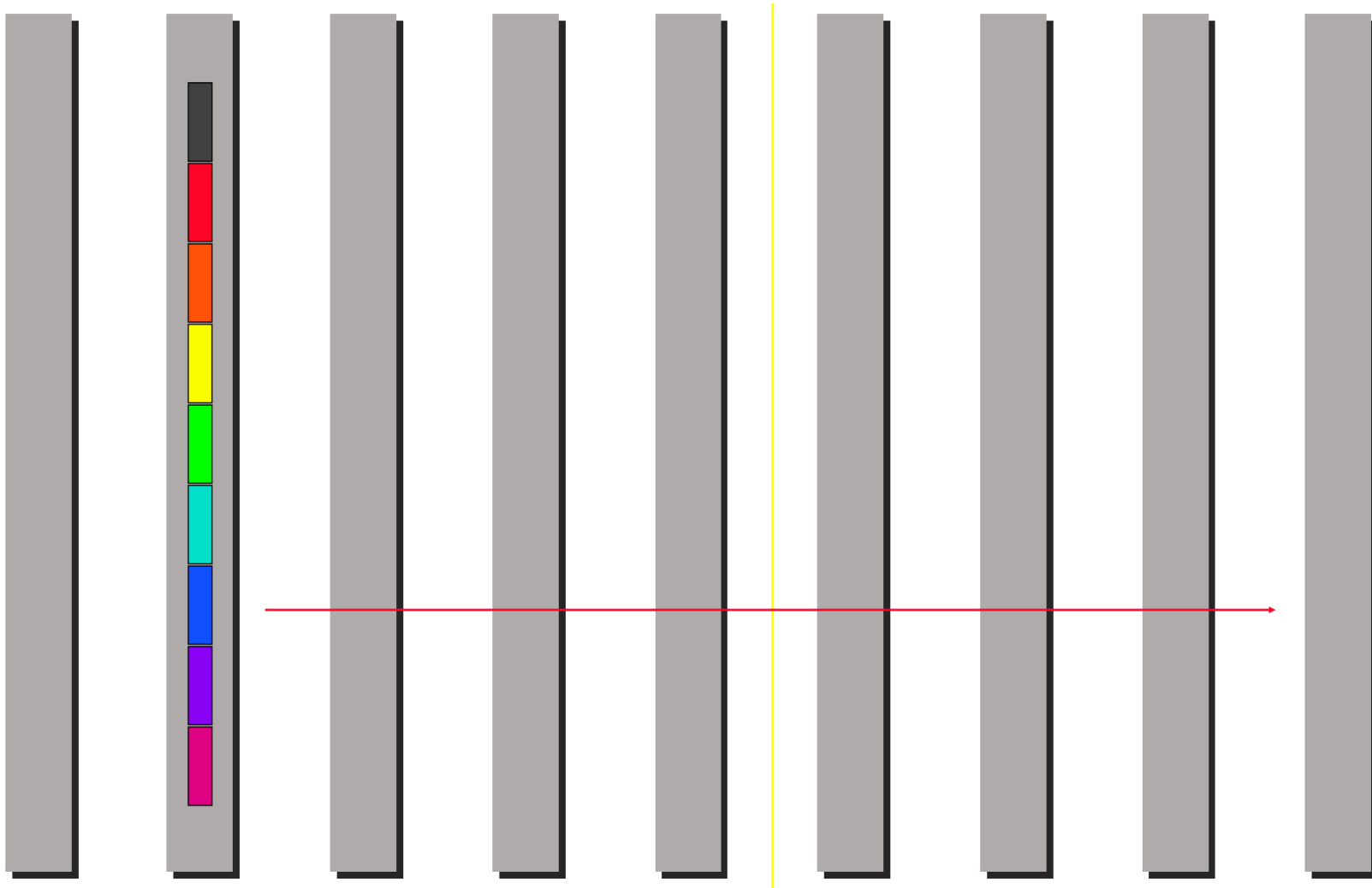
Before



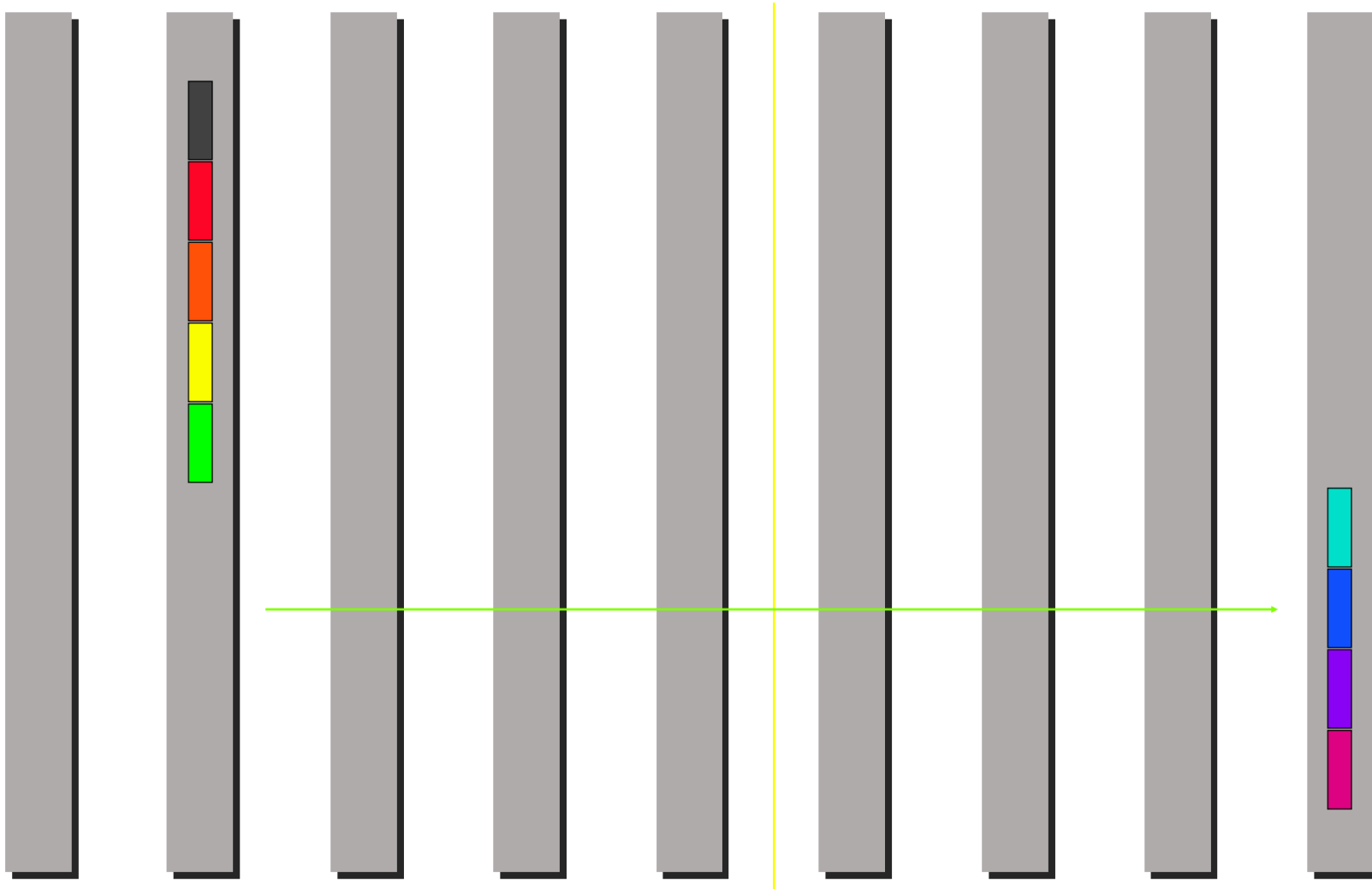
After



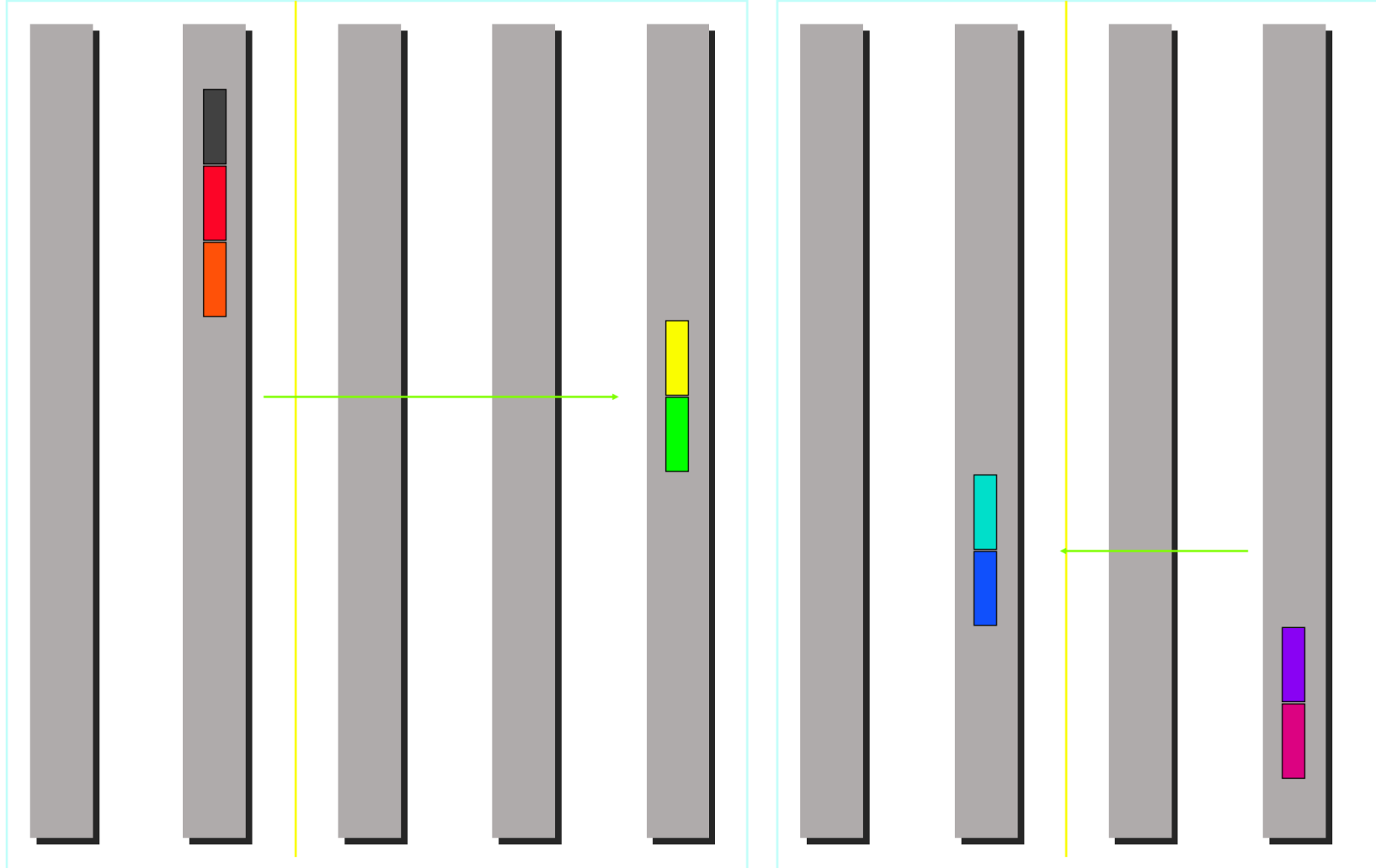
Scatter(1)



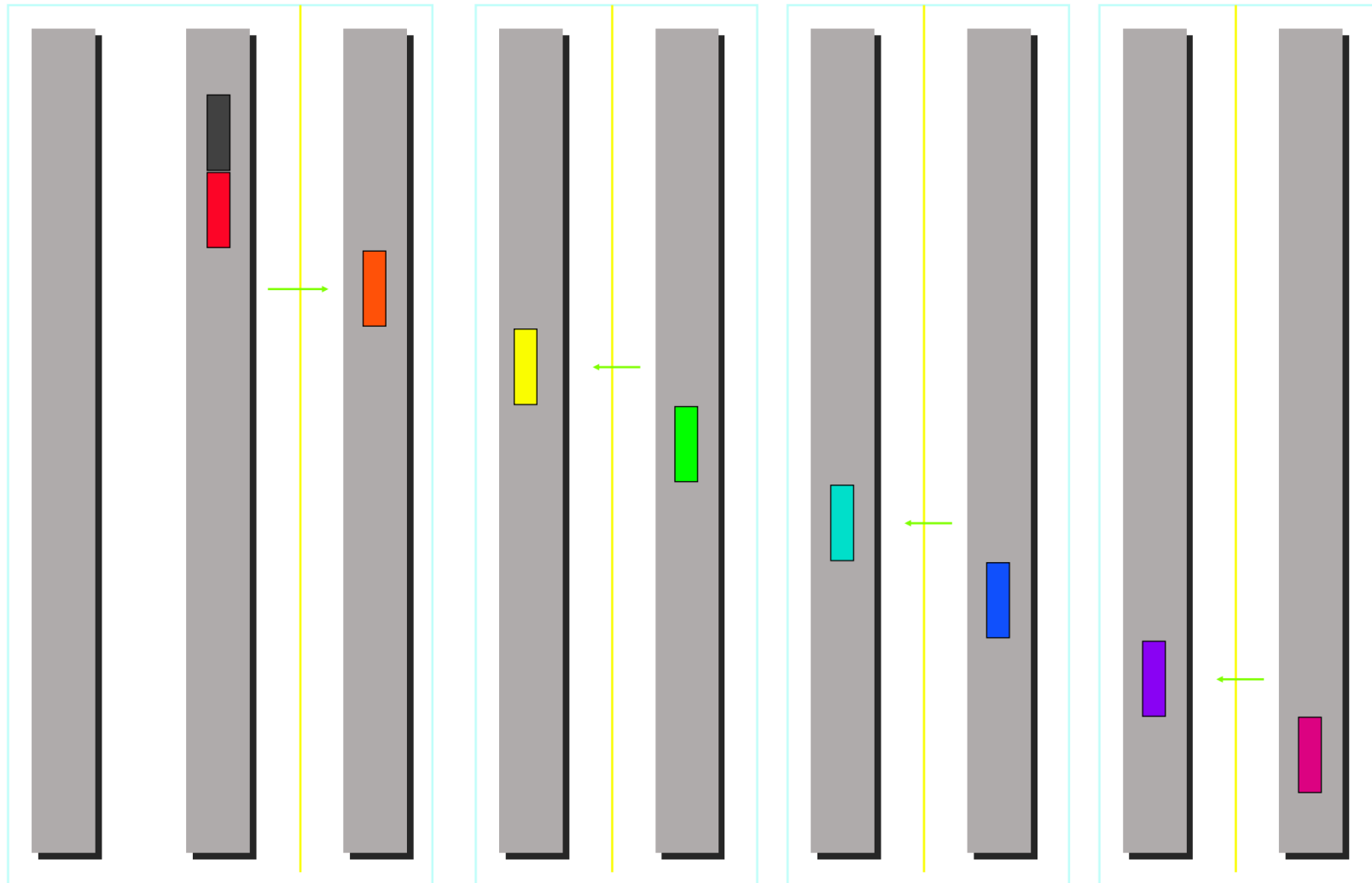
Scatter(2)



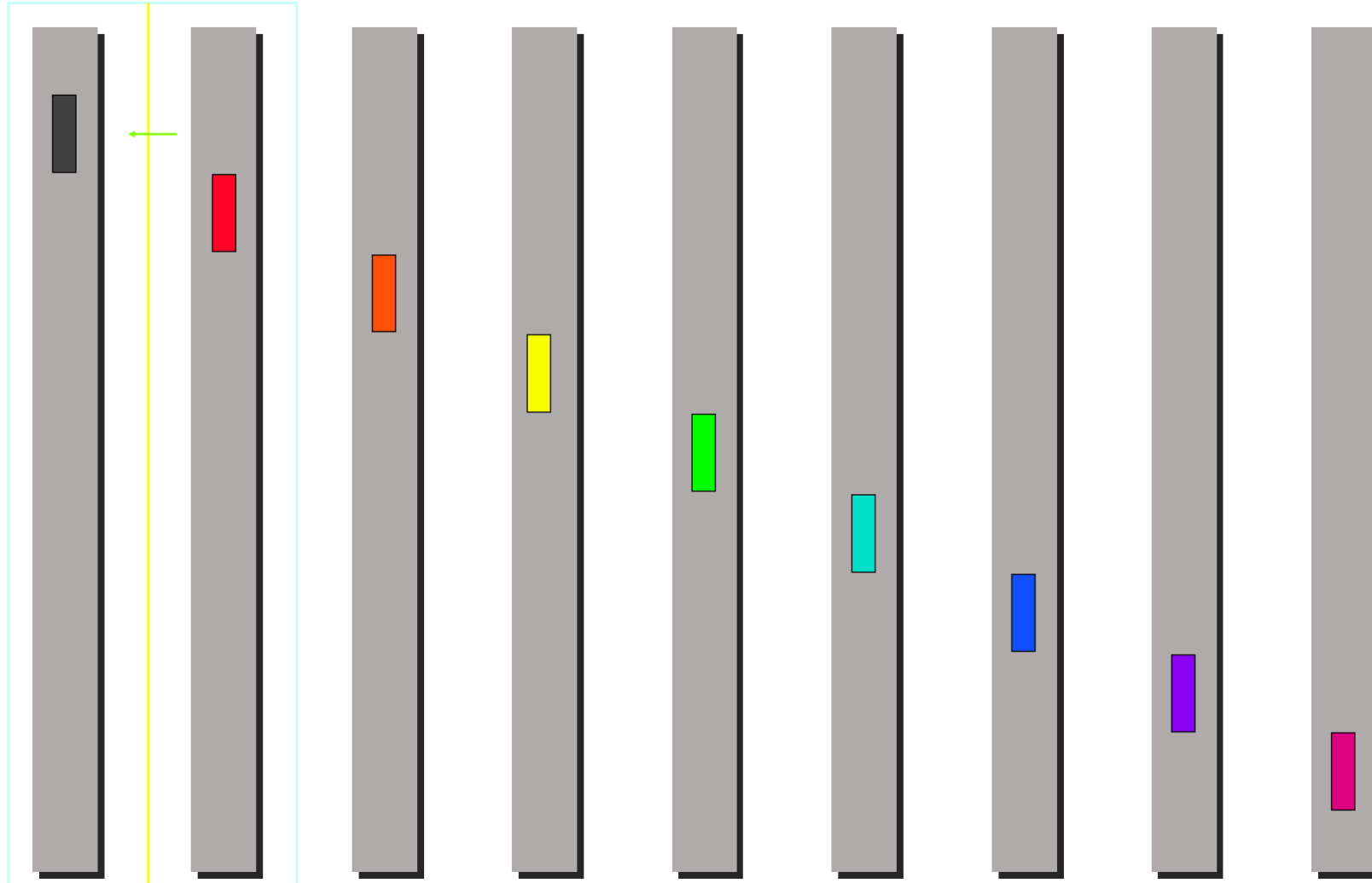
Scatter(3)



Scatter(4)



Scatter(5)



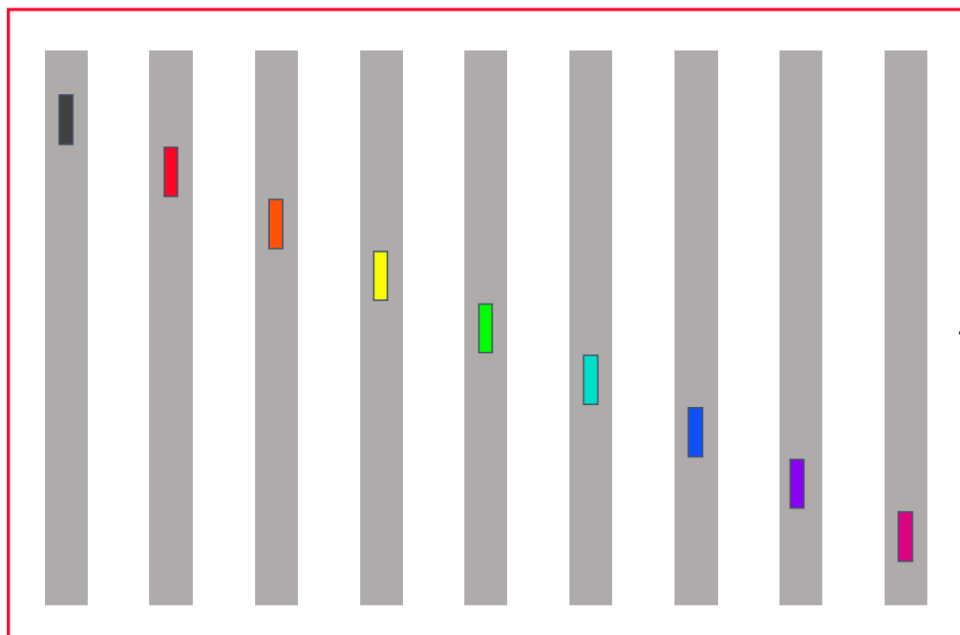
Cost of minimum spanning tree scatter

- Assumption: power of two number of nodes

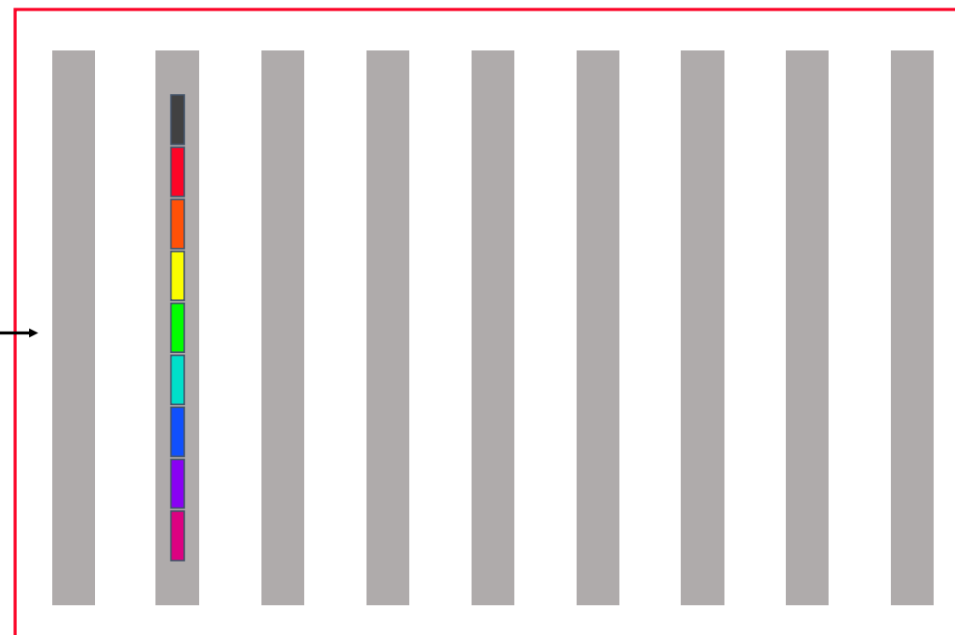
$$\sum_{k=1}^{\log(p)} \left(\alpha + \frac{n}{2^k} \beta \right) = \log(p) \alpha + \frac{p-1}{p} n \beta$$

Gather

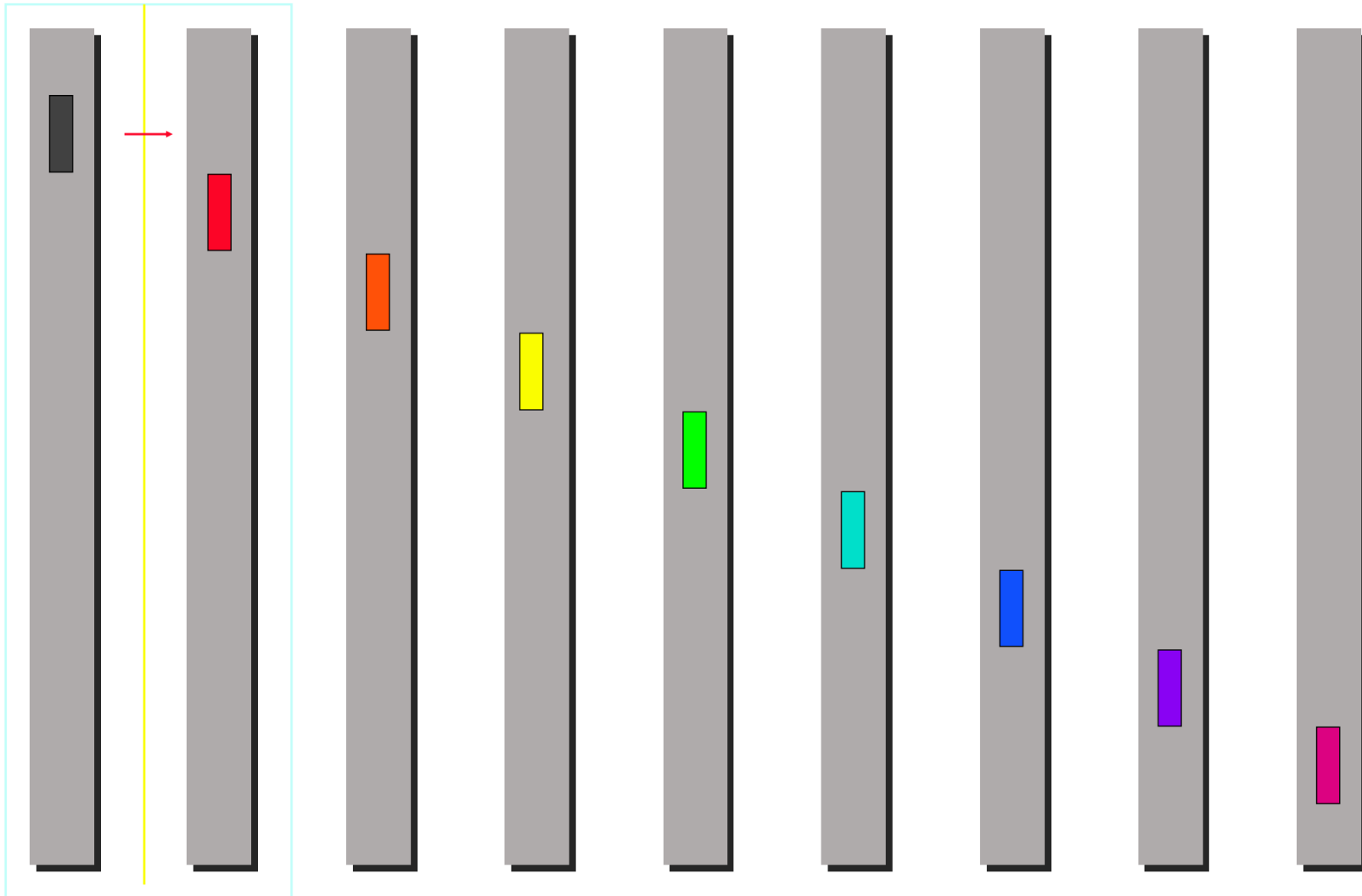
Before



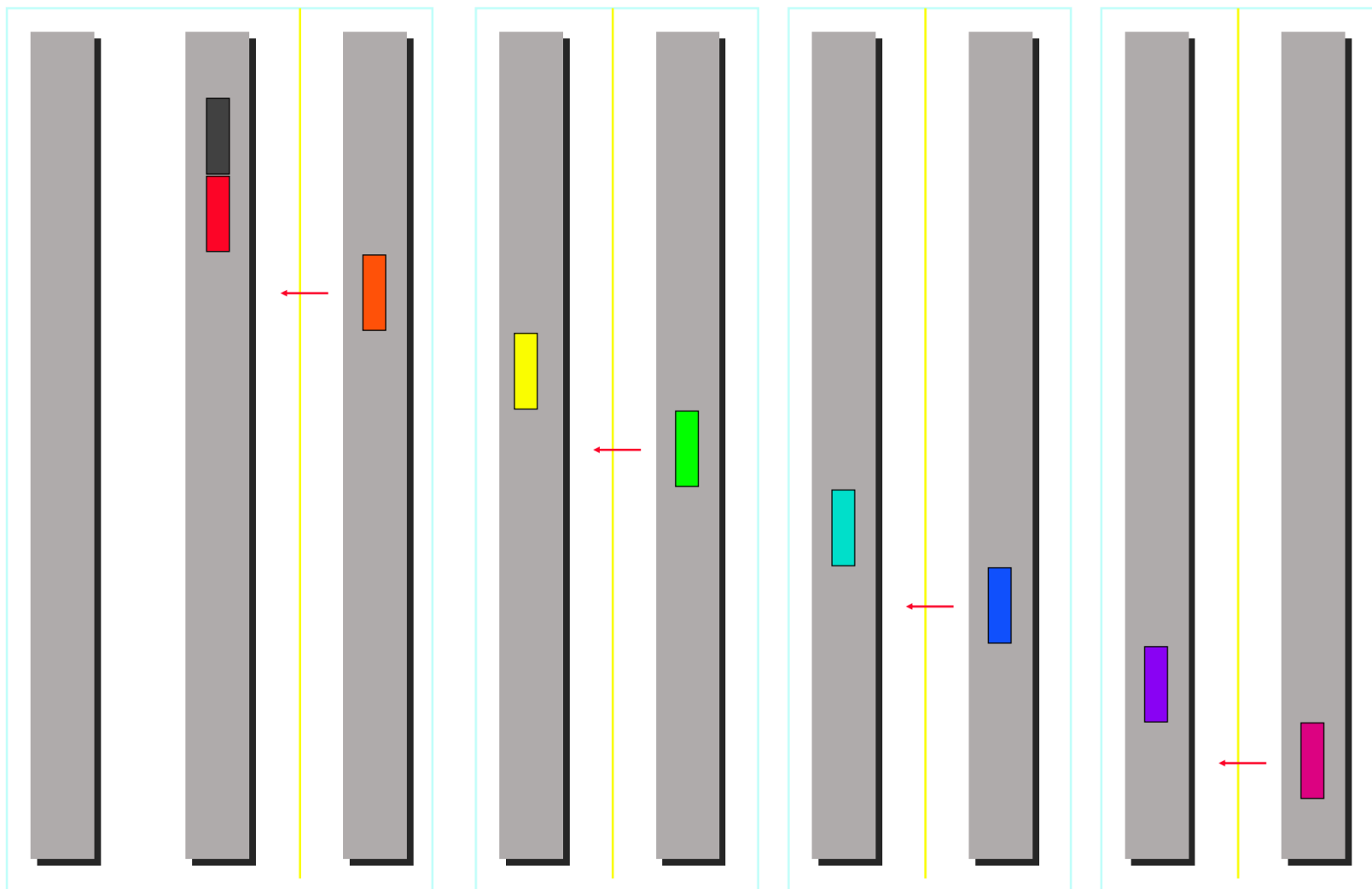
After



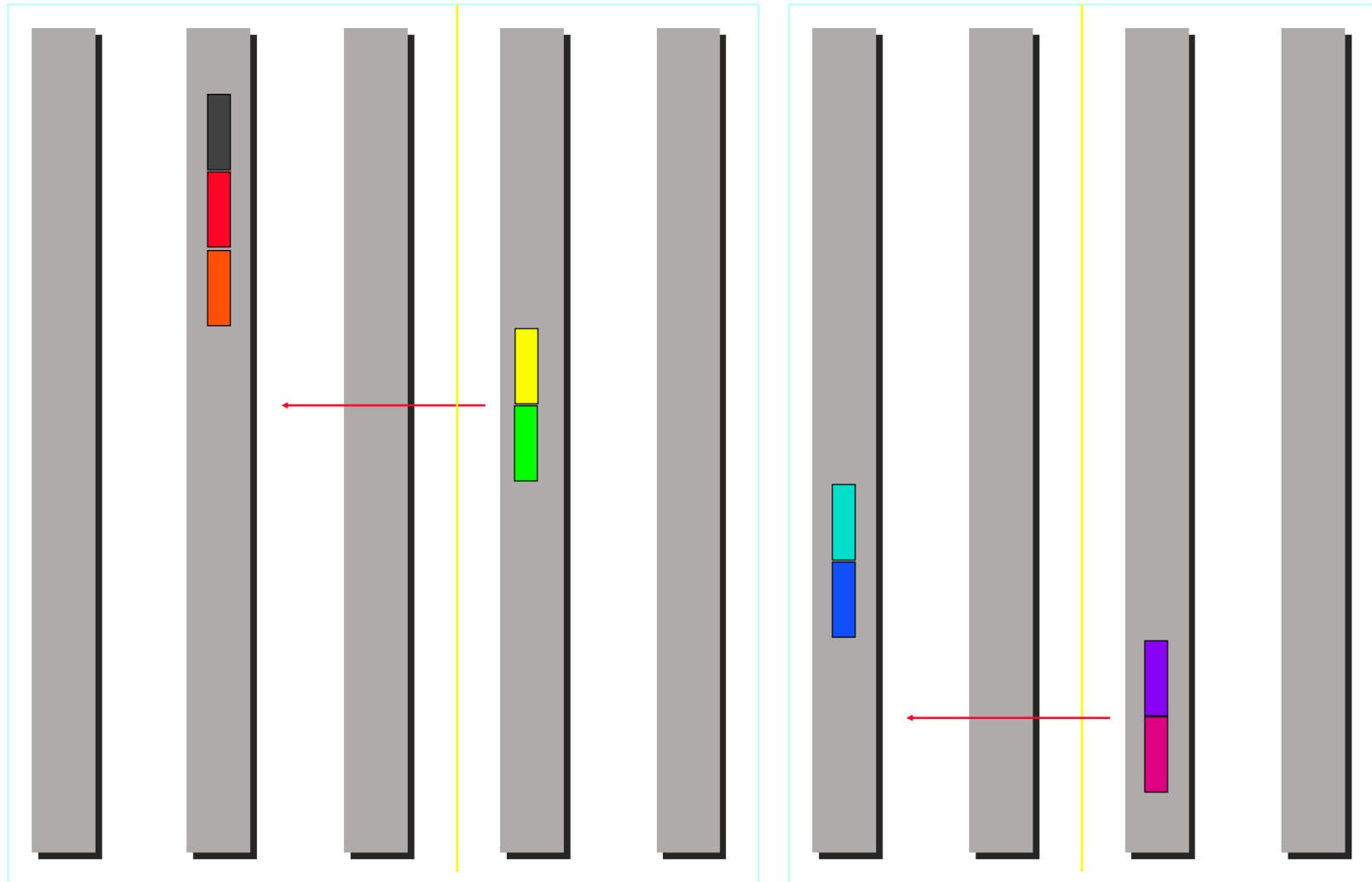
Gather(1)



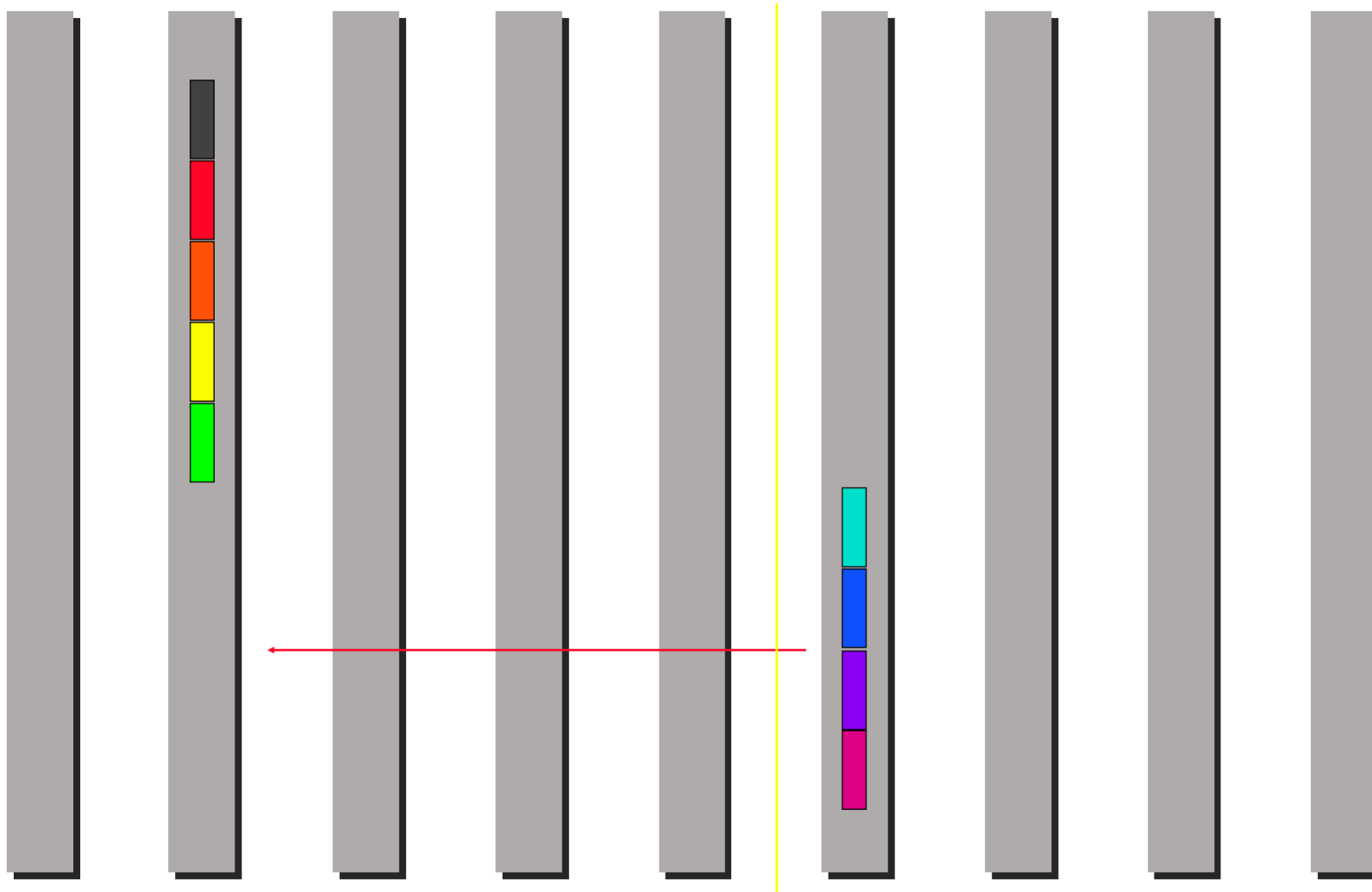
Gather(2)



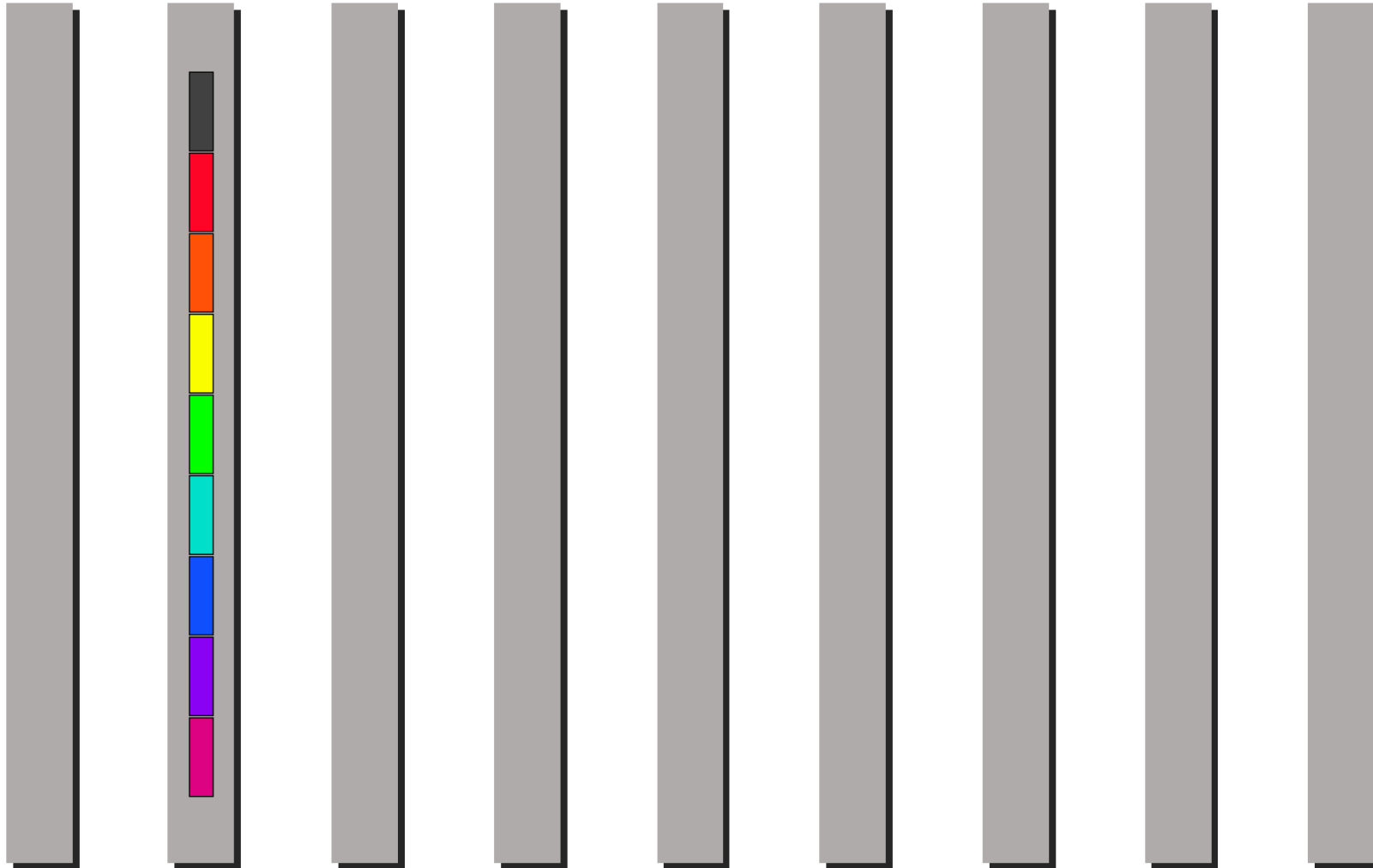
Gather(3)



Gather(4)



Gather(5)



Cost of minimum spanning tree gather

- Assumption: power of two number of nodes

$$\sum_{k=1}^{\log(p)} \left(\alpha + \frac{n}{2^k} \beta \right)$$

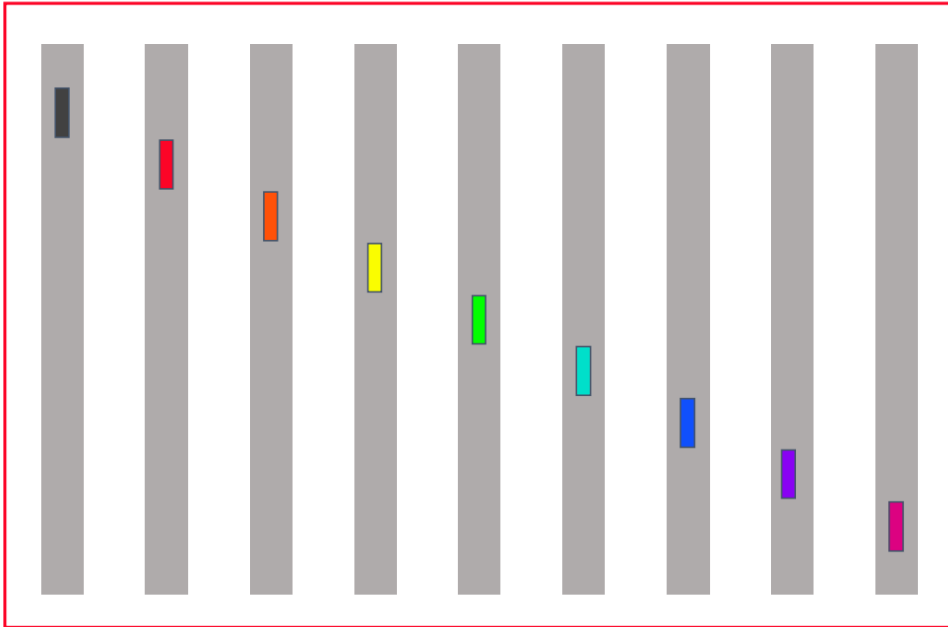
=

$$\log(p) \alpha + \frac{p-1}{p} n \beta$$

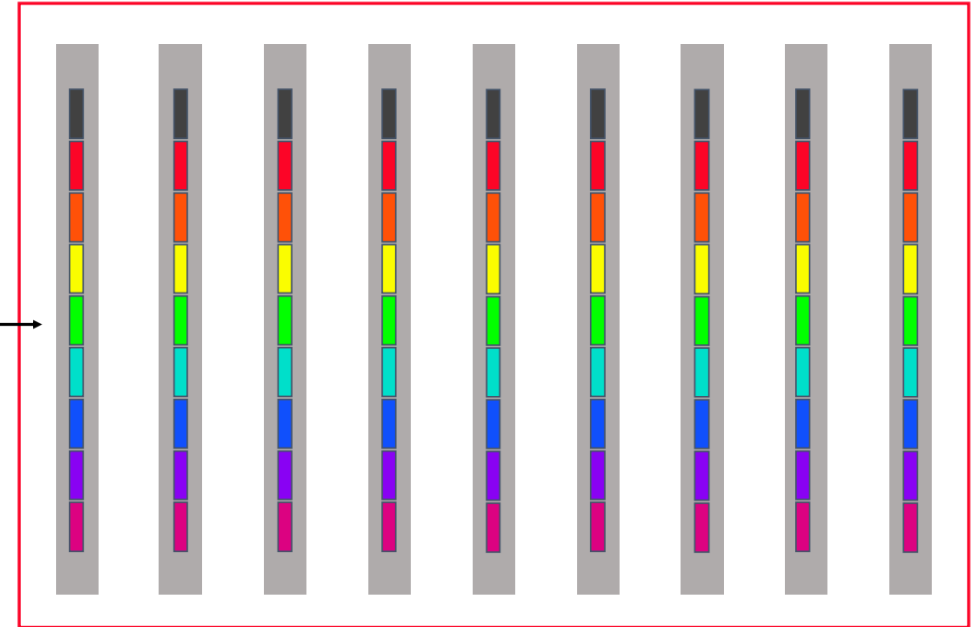
Same with scatter

Allgather

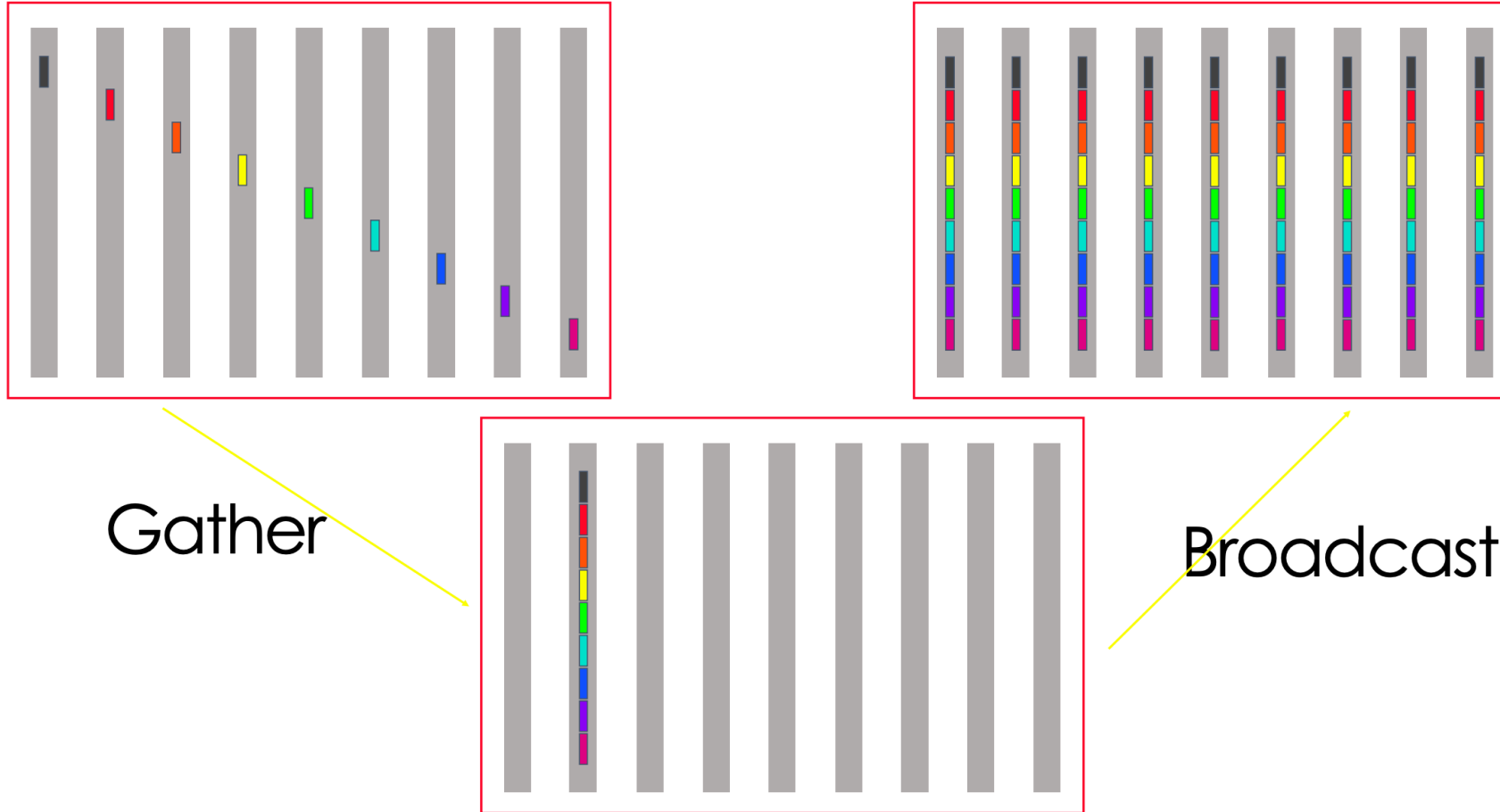
Before



After



Allgather



Cost of gather/broadcast allgather

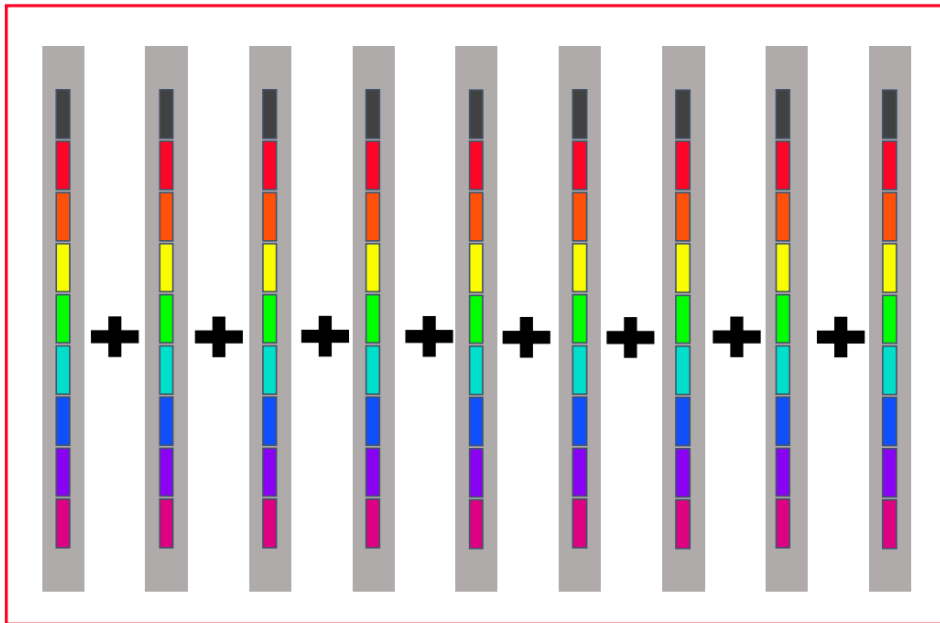
- Assumption: power of two number of nodes

gather $\log(p)\alpha + \frac{p-1}{p}n\beta$

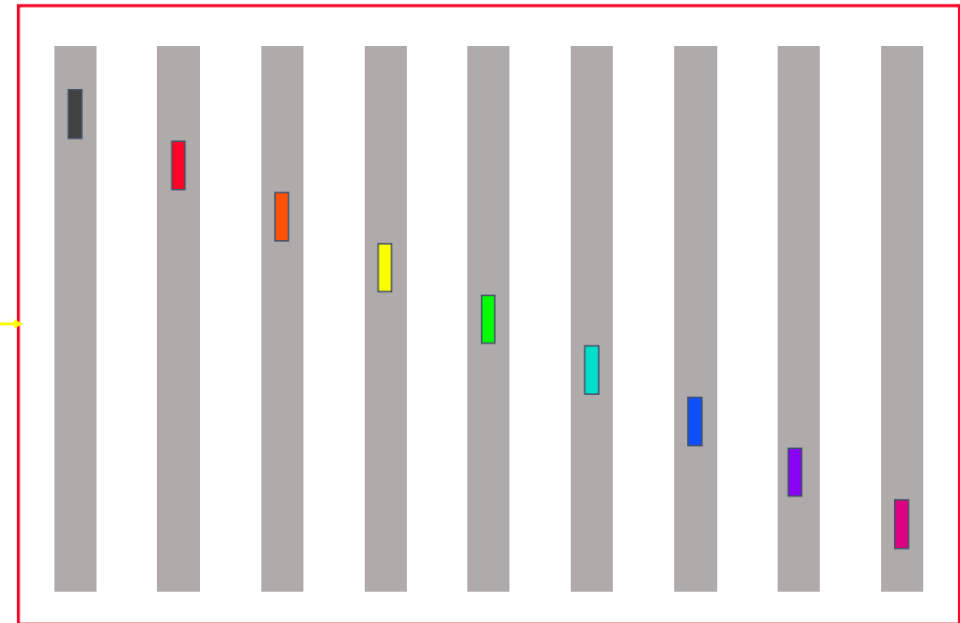
broadcast $\frac{\log(p)(\alpha + n\beta)}{2\log(p)\alpha + \left(\frac{p-1}{p} + \log(p)\right)n\beta}$

Reduce-scatter

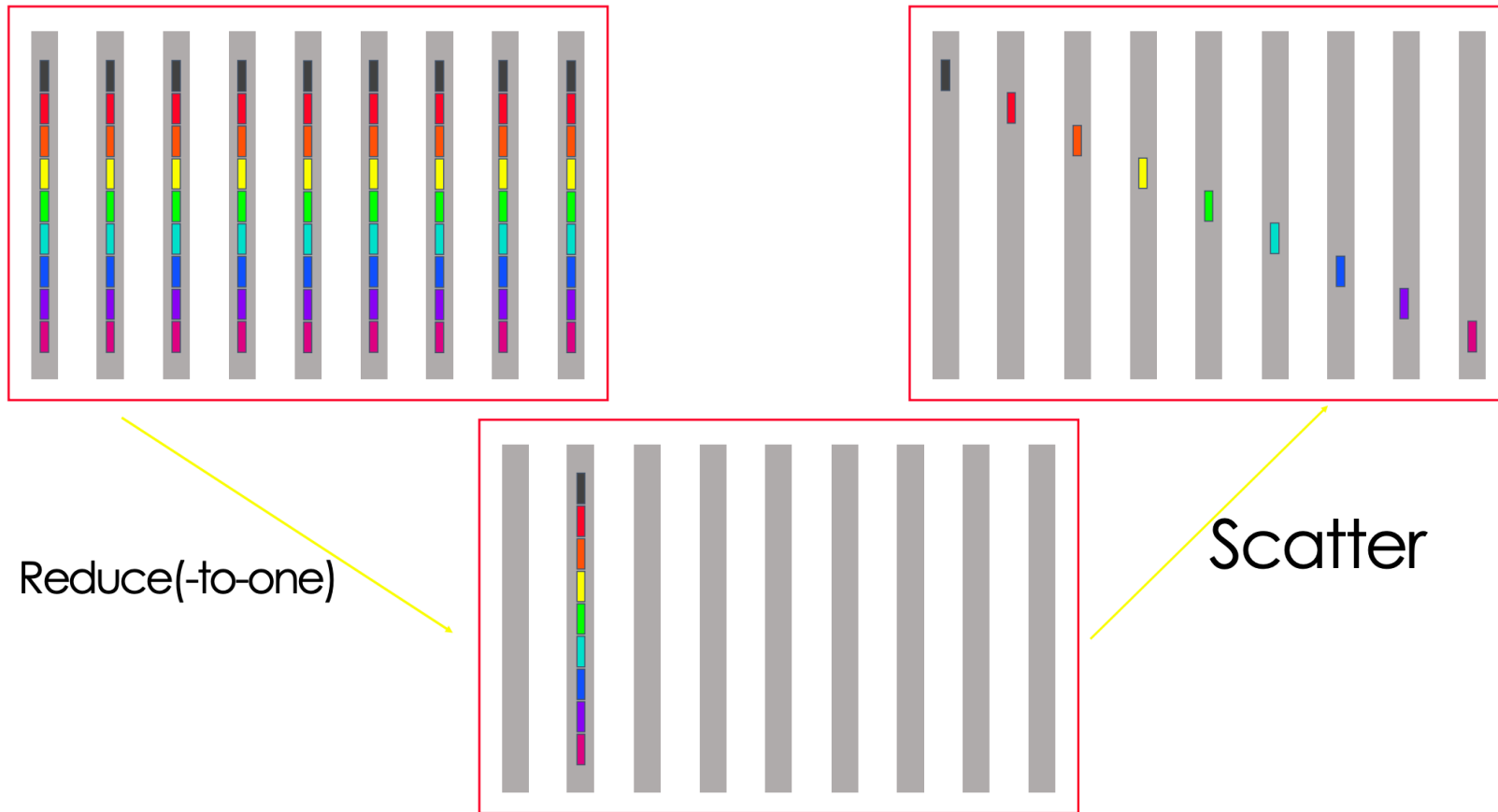
Before



After



Reduce-scatter



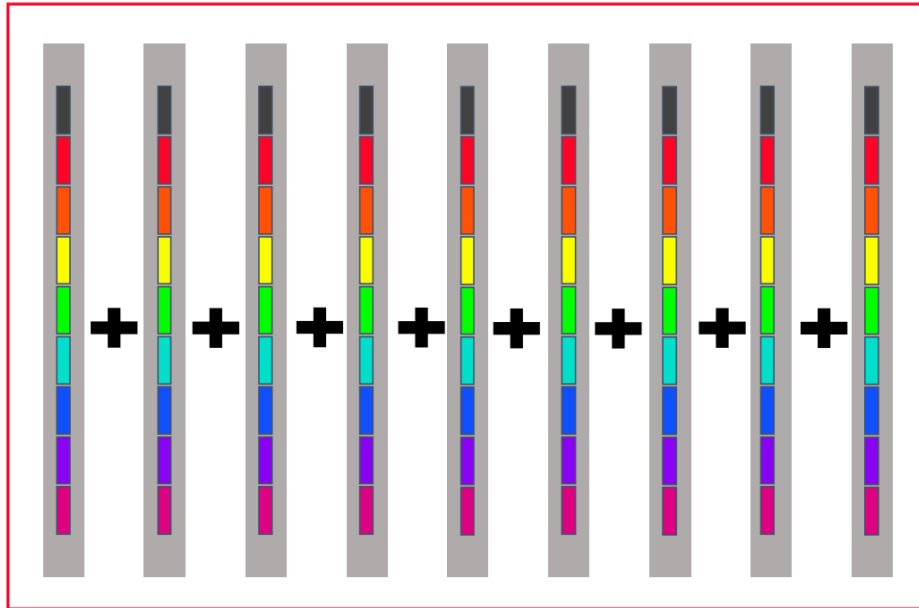
Cost of reduce-scatter

- Assumption: power of two number of nodes

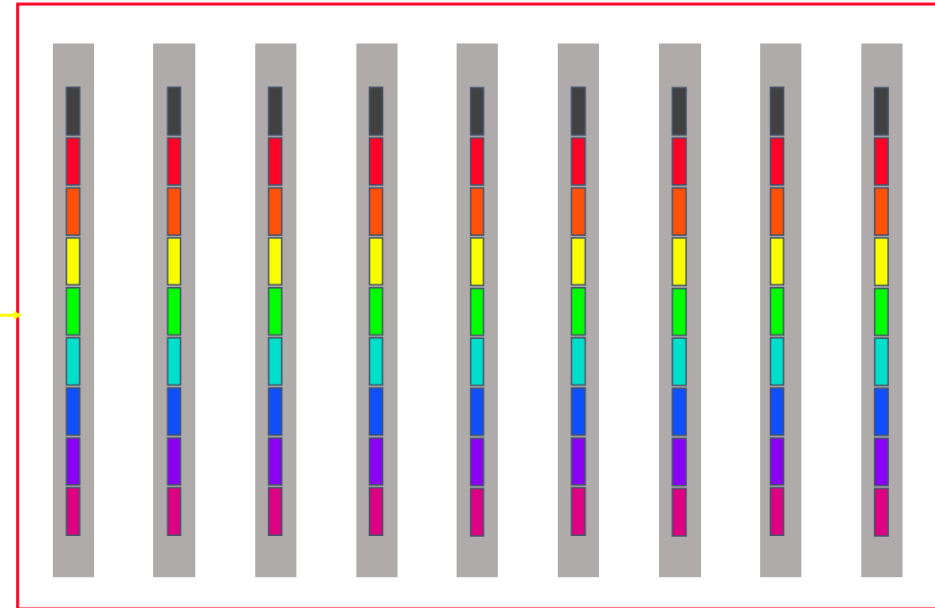
$$\begin{array}{l} \text{Reduce(-to-one)} \quad \log(p)(\alpha + n\beta + n\gamma) \\ \text{scatter} \quad \log(p)\alpha + \frac{p-1}{p}n\beta \\ \hline 2\log(p)\alpha + \left(\frac{p-1}{p} + \log(p)\right)n\beta + \log(p)n\gamma \end{array}$$

Allreduce

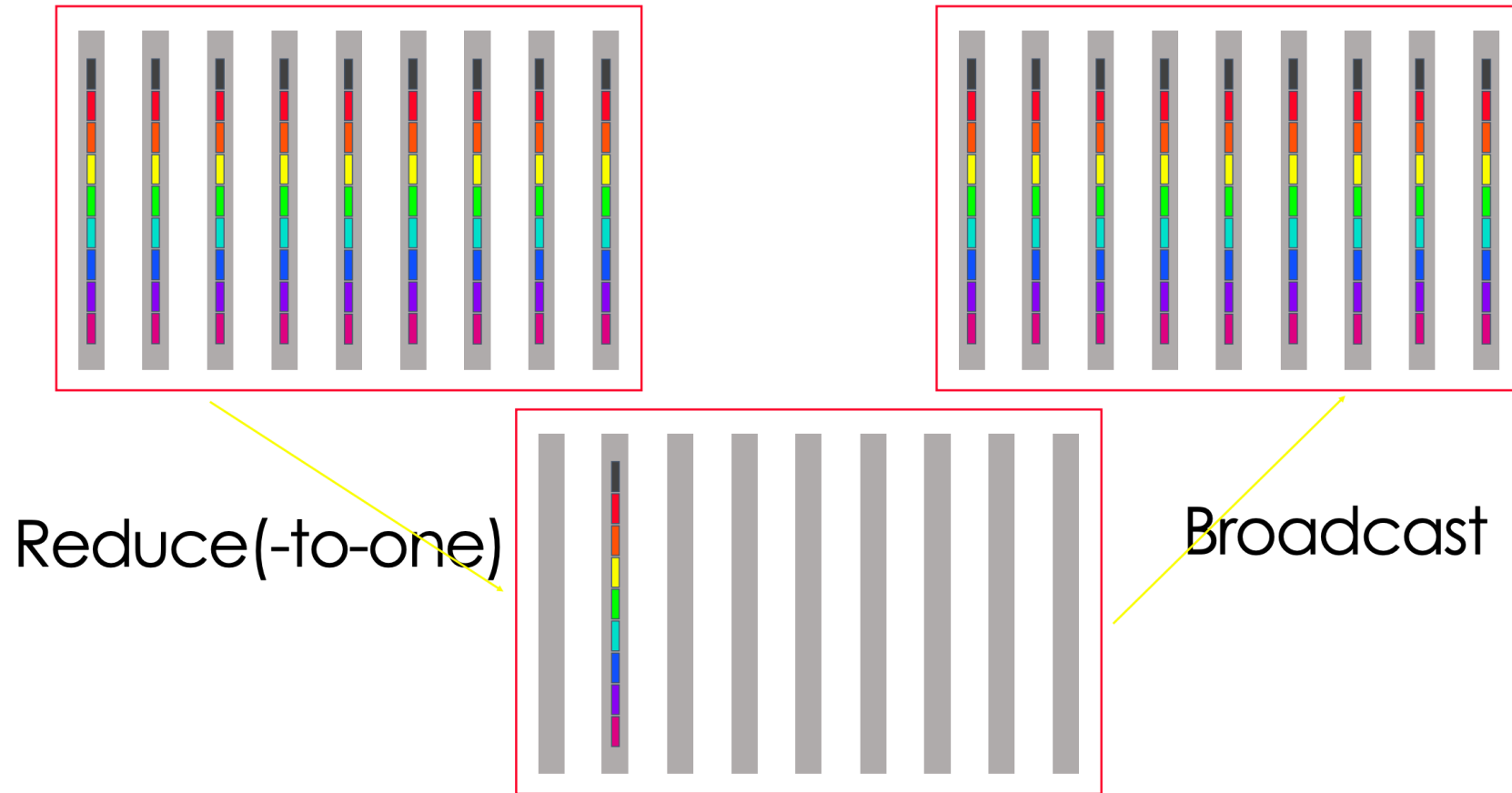
Before



After



Allreduce



Cost of reduce/broadcast Allreduce

- Assumption: power of two number of nodes

$$\text{Reduce(-to-one)} \log(p)(\alpha + n\beta + n\gamma)$$

$$\frac{\text{broadcast} \log(p)(\alpha + n\beta)}{2\log(p)\alpha + 2\log(p)n\beta + \log(p)n\gamma}$$

Summary of MST algorithms

Reduce(-to-one)

$$\log(p)(\alpha + n\beta + n\gamma)$$

Scatter

$$\log(p)\alpha + \frac{p-1}{p}n\beta$$

Gather

$$\log(p)\alpha + \frac{p-1}{p}n\beta$$

Broadcast

$$\log(p)(\alpha + n\beta)$$

Reduce-scatter

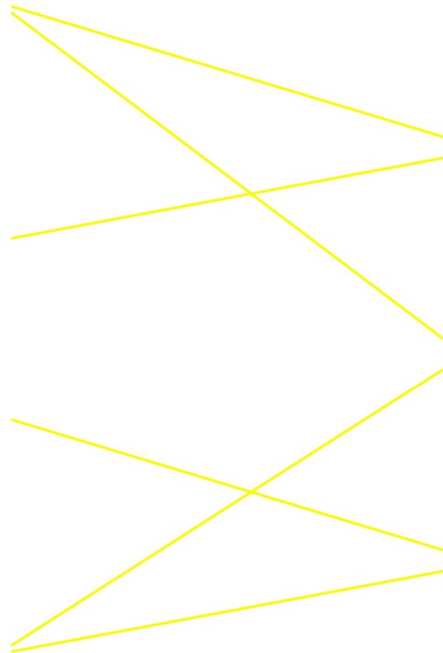
$$2\log(p)\alpha + \log(p)n(\beta + \gamma) + \frac{p-1}{p}n\beta$$

Allreduce

$$2\log(p)\alpha + \log(p)n(2\beta + \gamma)$$

Allgather

$$2\log(p)\alpha + \log(p)n\beta + \frac{p-1}{p}n\beta$$



Pros and cons of MST algorithms

- Emphasize low latency
 - MST-based algorithm is latency-optimal
- Problem of Minimum Spanning Tree Algorithm?
 - It prioritizes latency rather than bandwidth
 - Some links are idle

Large message

- Communication Model

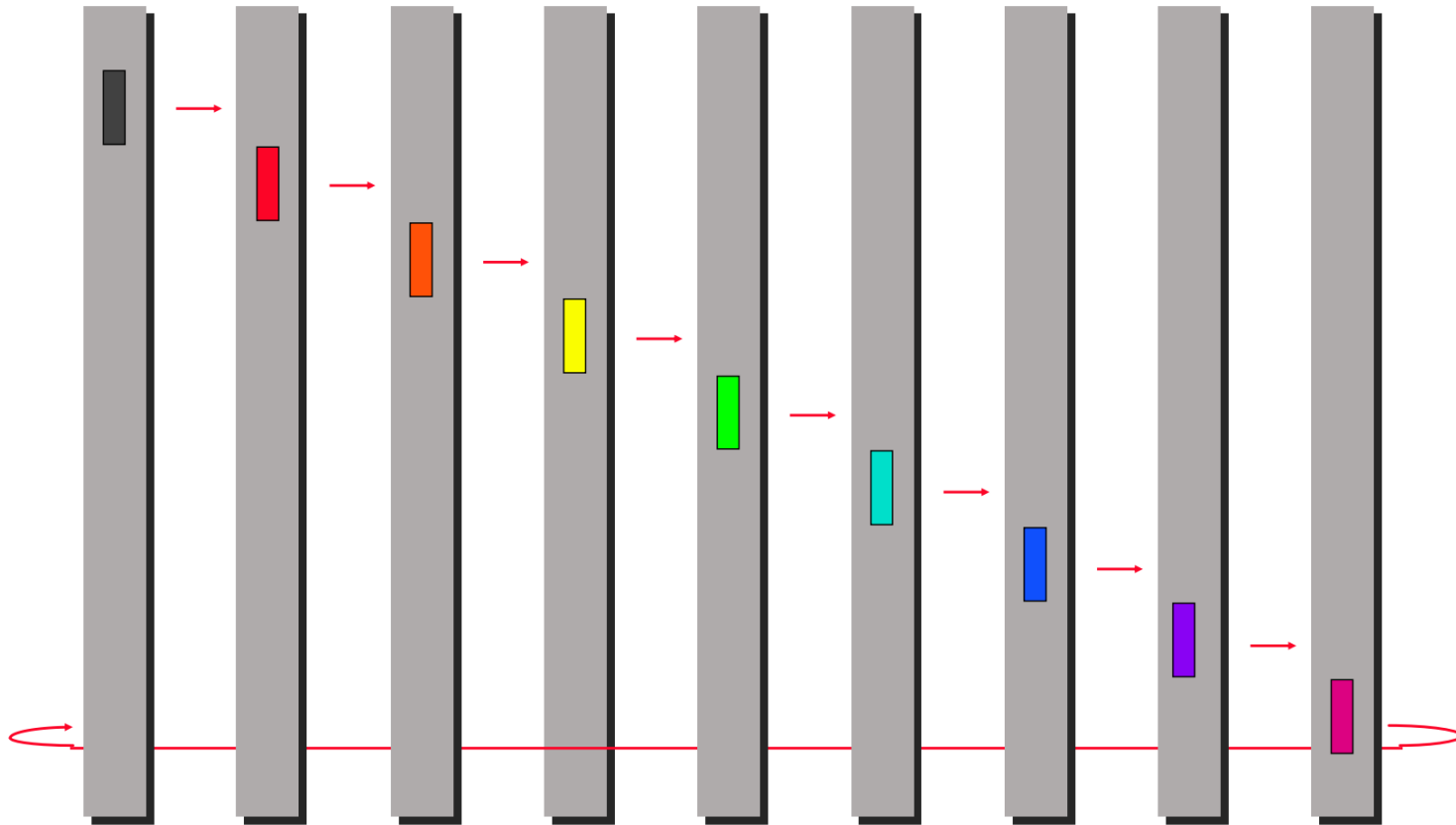
$$\alpha + n \beta, \beta = \frac{1}{B}$$

- The second term dominates $\rightarrow$ minimize the second term
 - Utilize the bandwidth as much as possible

General principles for large messages

- Use all the links between every two nodes
 - Minimal cost
 - Implementation for arbitrary numbers of nodes
 - no network conflicts
- A logical ring can be embedded in a physical linear array with worm-hole routing, since the “wrap-around” message doesn’t conflict

General principles for large messages

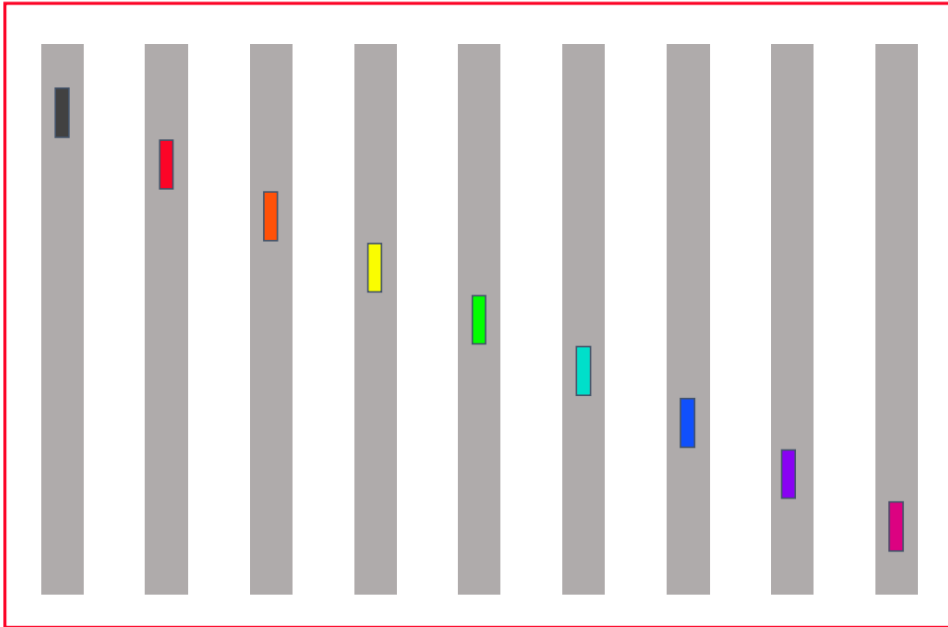


Benefits of ring algorithm

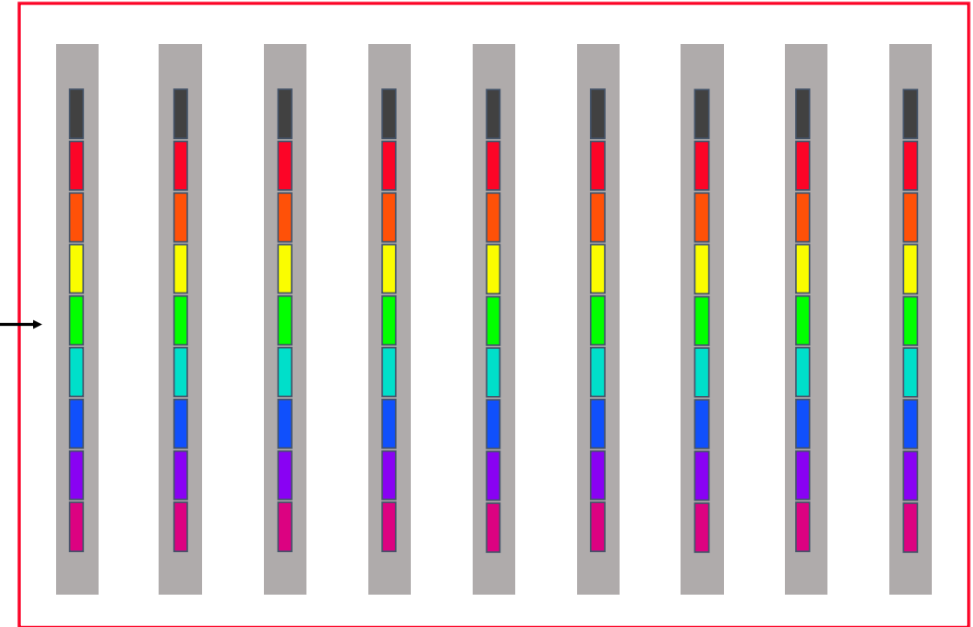
- Fully utilize the bandwidth (bandwidth optimal)
- Arbitrary numbers of node

Allgather

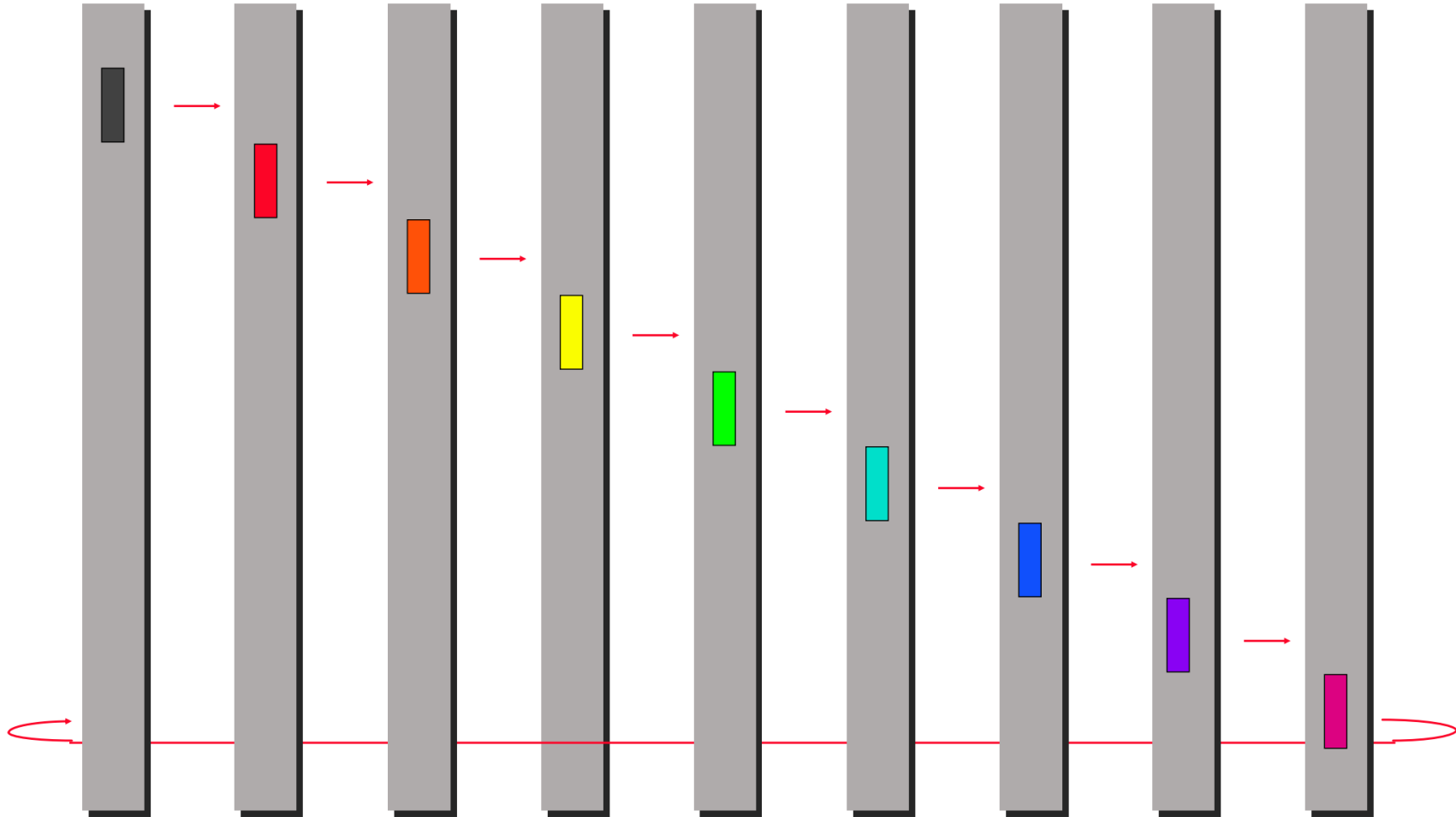
Before



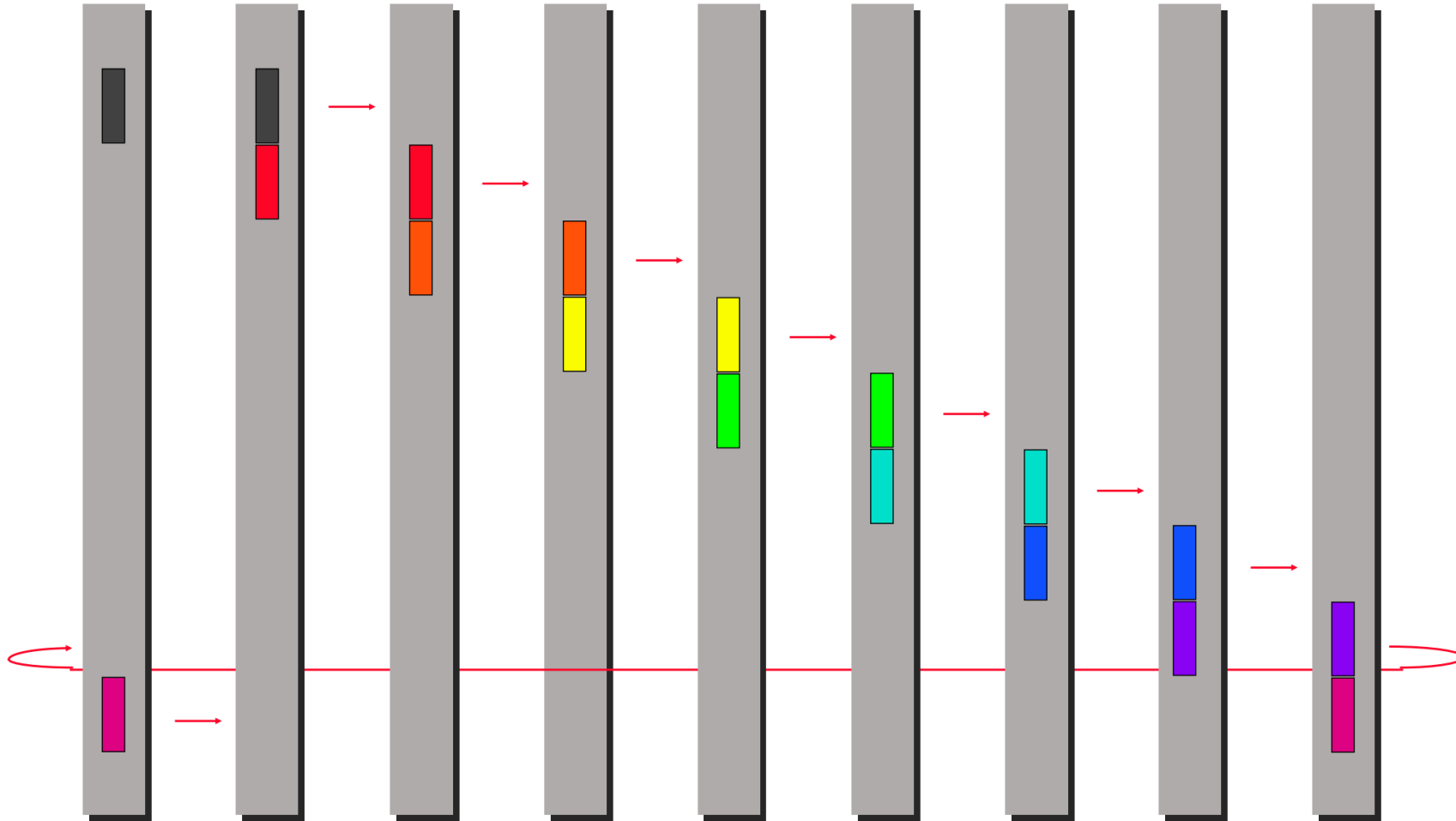
After



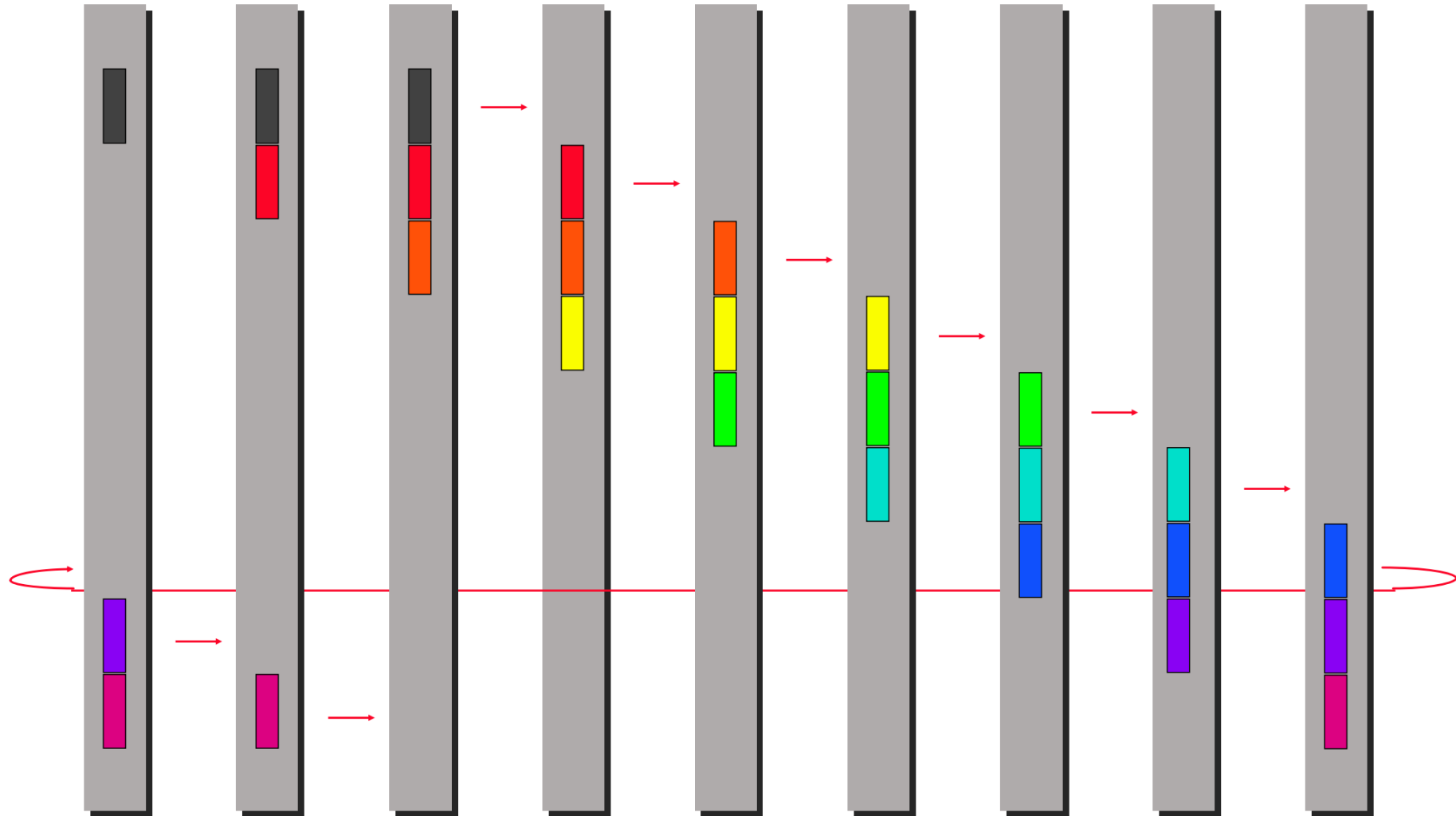
Allgather(1)



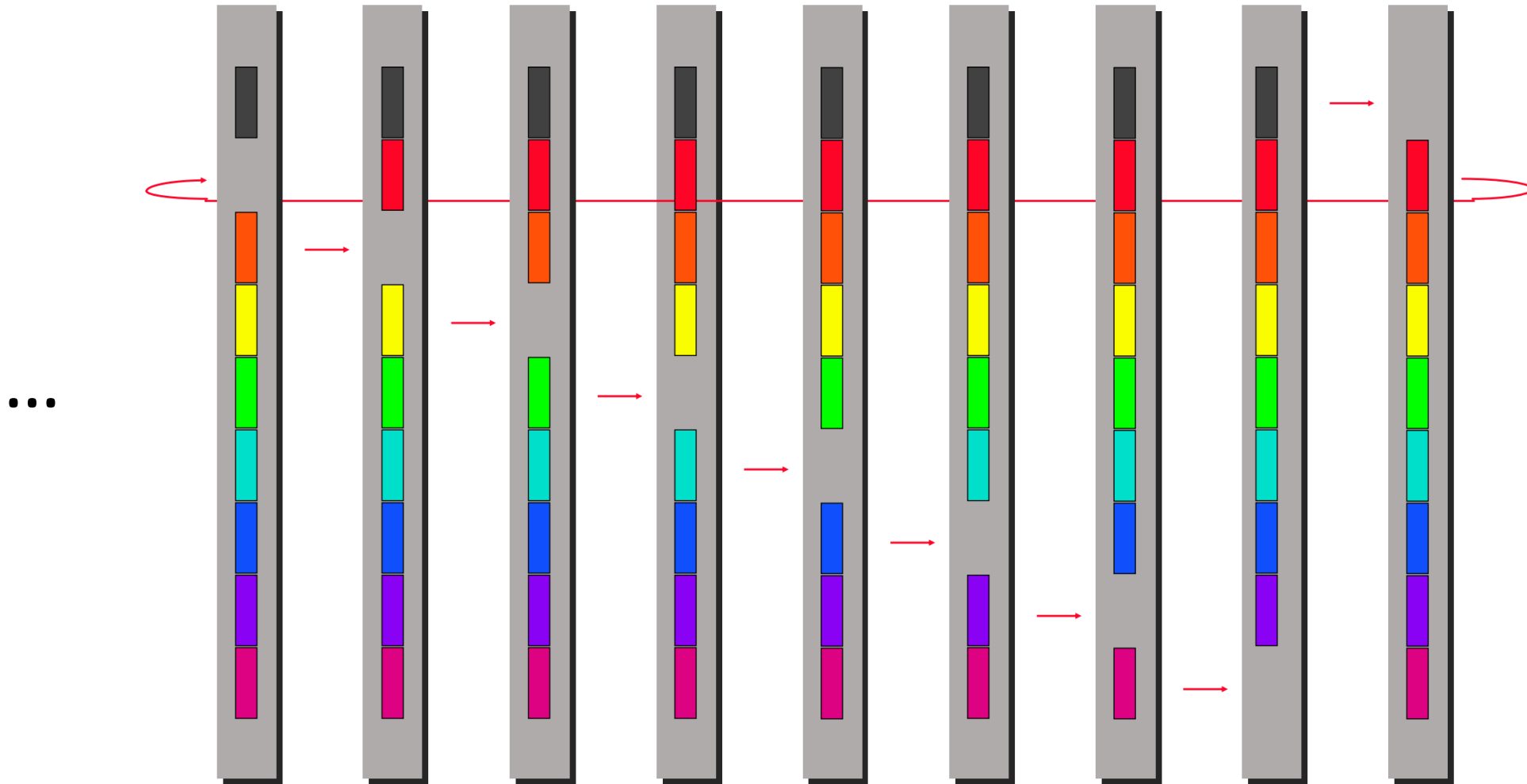
Allgather(2)



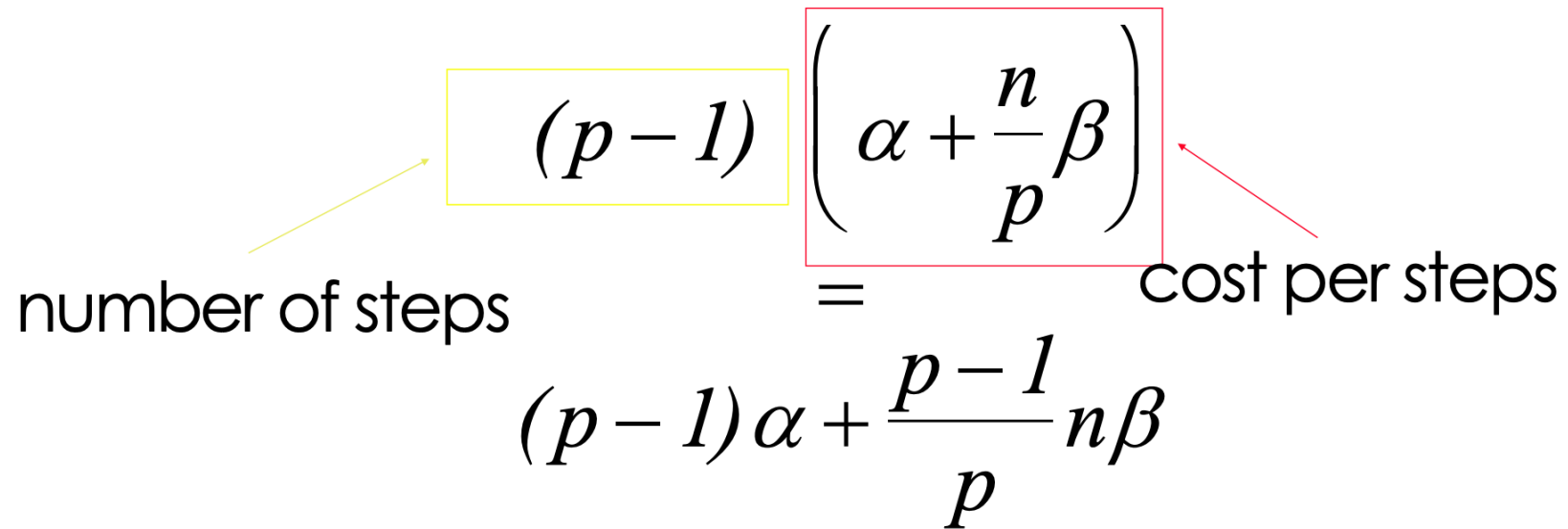
Allgather(3)



Allgather(4)



Cost of Allgather



The diagram illustrates the cost of an Allgather operation. It features a mathematical expression with two parts highlighted by colored boxes and arrows. A yellow box highlights the term $(p-1)$, with a yellow arrow pointing to it from the text "number of steps". A red box highlights the term $\left(\alpha + \frac{n}{p}\beta\right)$, with a red arrow pointing to it from the text "cost per steps". Below these, an equals sign is followed by the expanded formula $(p-1)\alpha + \frac{p-1}{p}n\beta$.

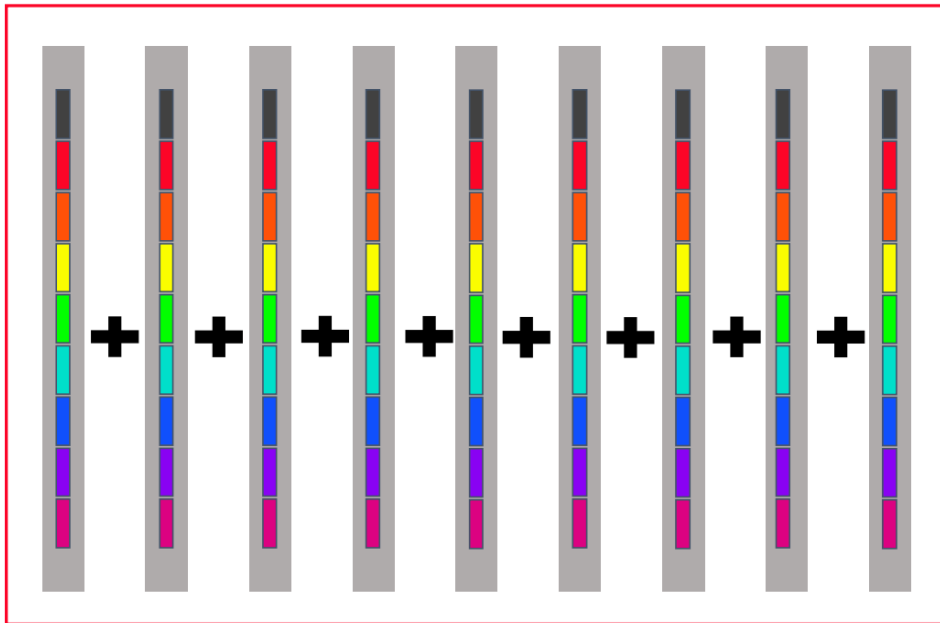
$$(p-1) \left(\alpha + \frac{n}{p}\beta \right) = (p-1)\alpha + \frac{p-1}{p}n\beta$$

number of steps

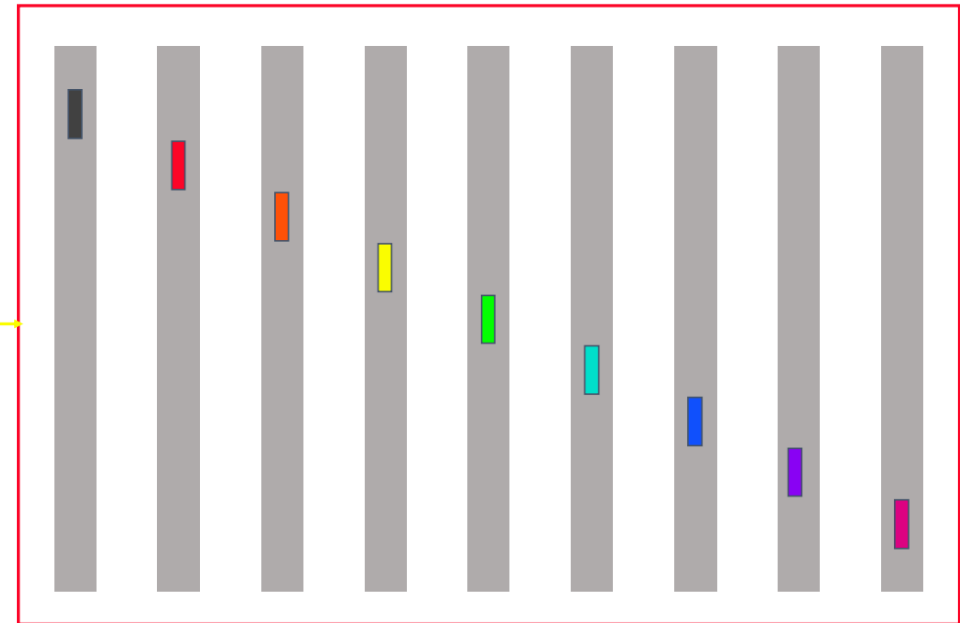
cost per steps

Reduce-scatter

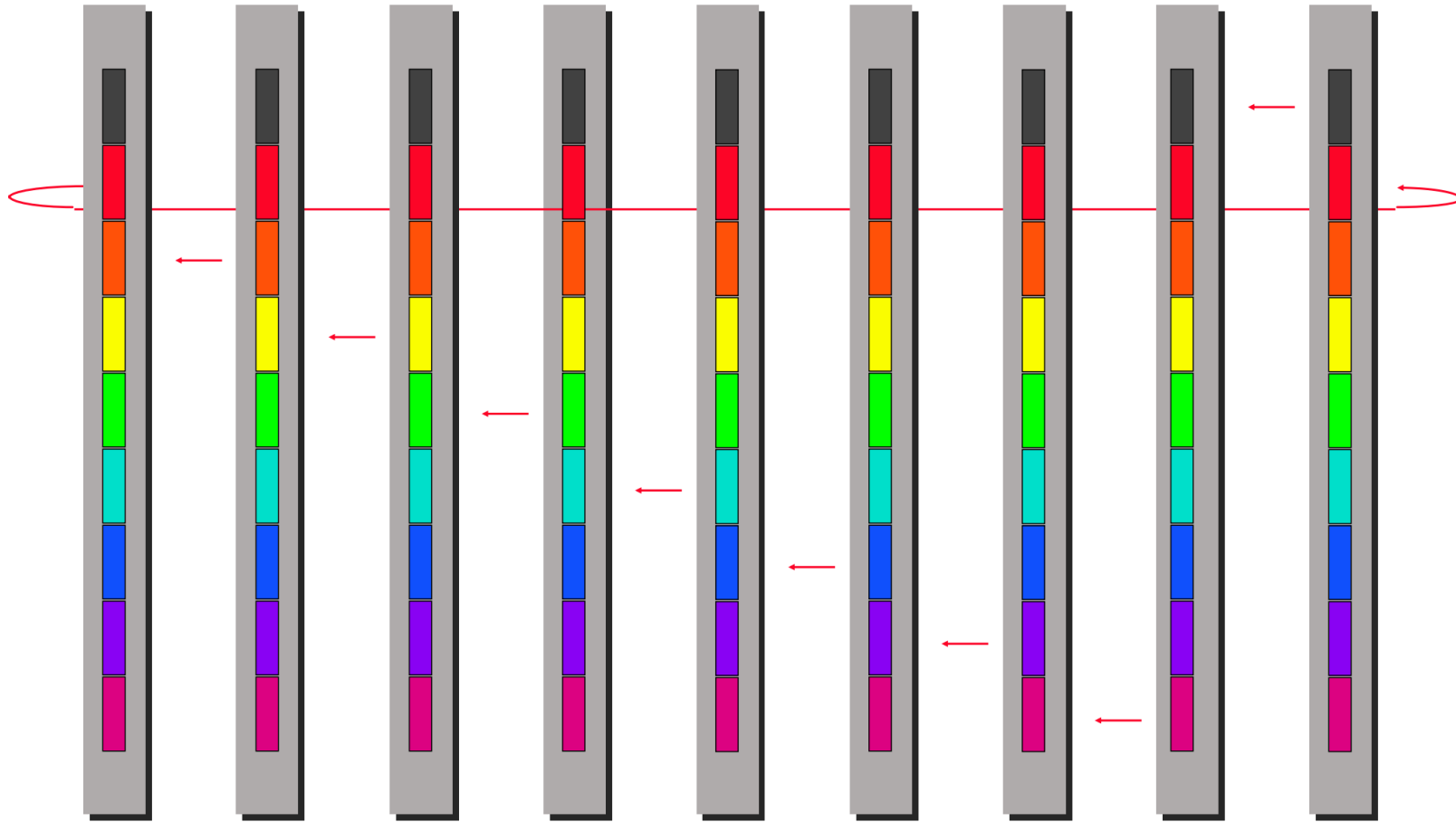
Before



After

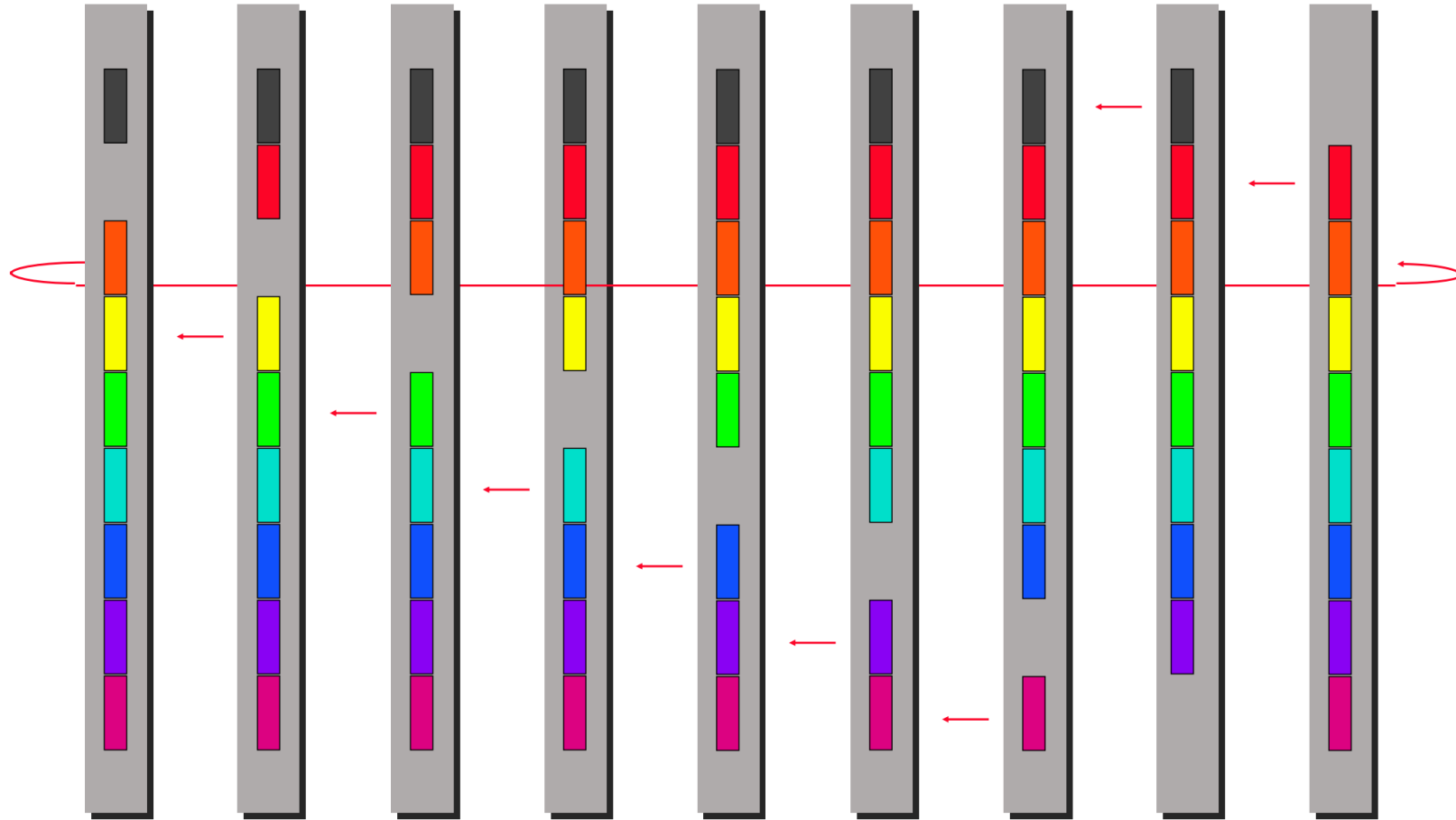


Reduce-scatter(1)



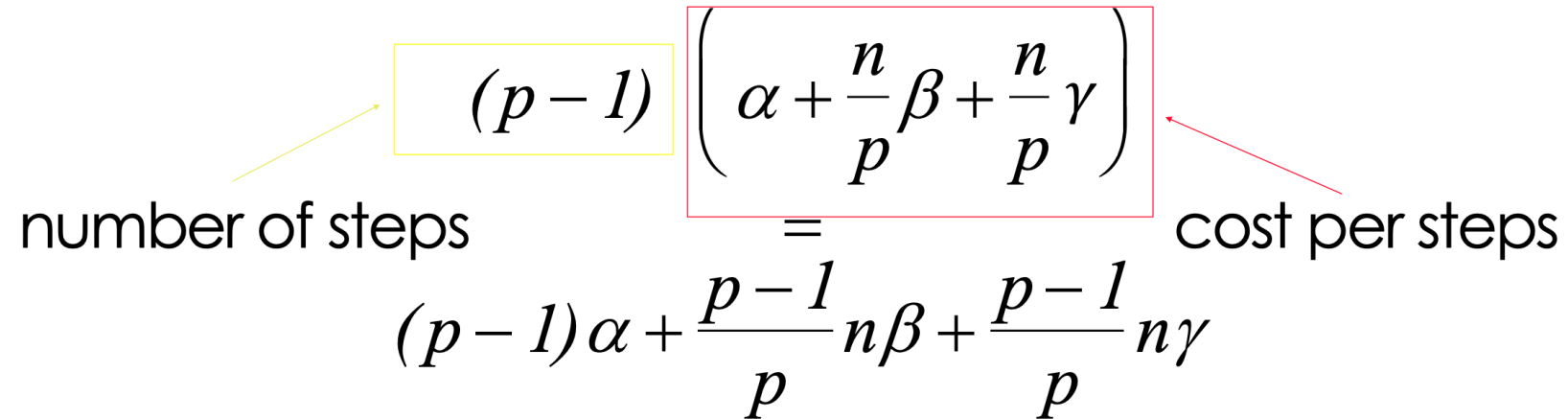
Reversed ring!

Reduce-scatter(2)



Complete the remaining steps yourself

Cost of Reduce-scatter



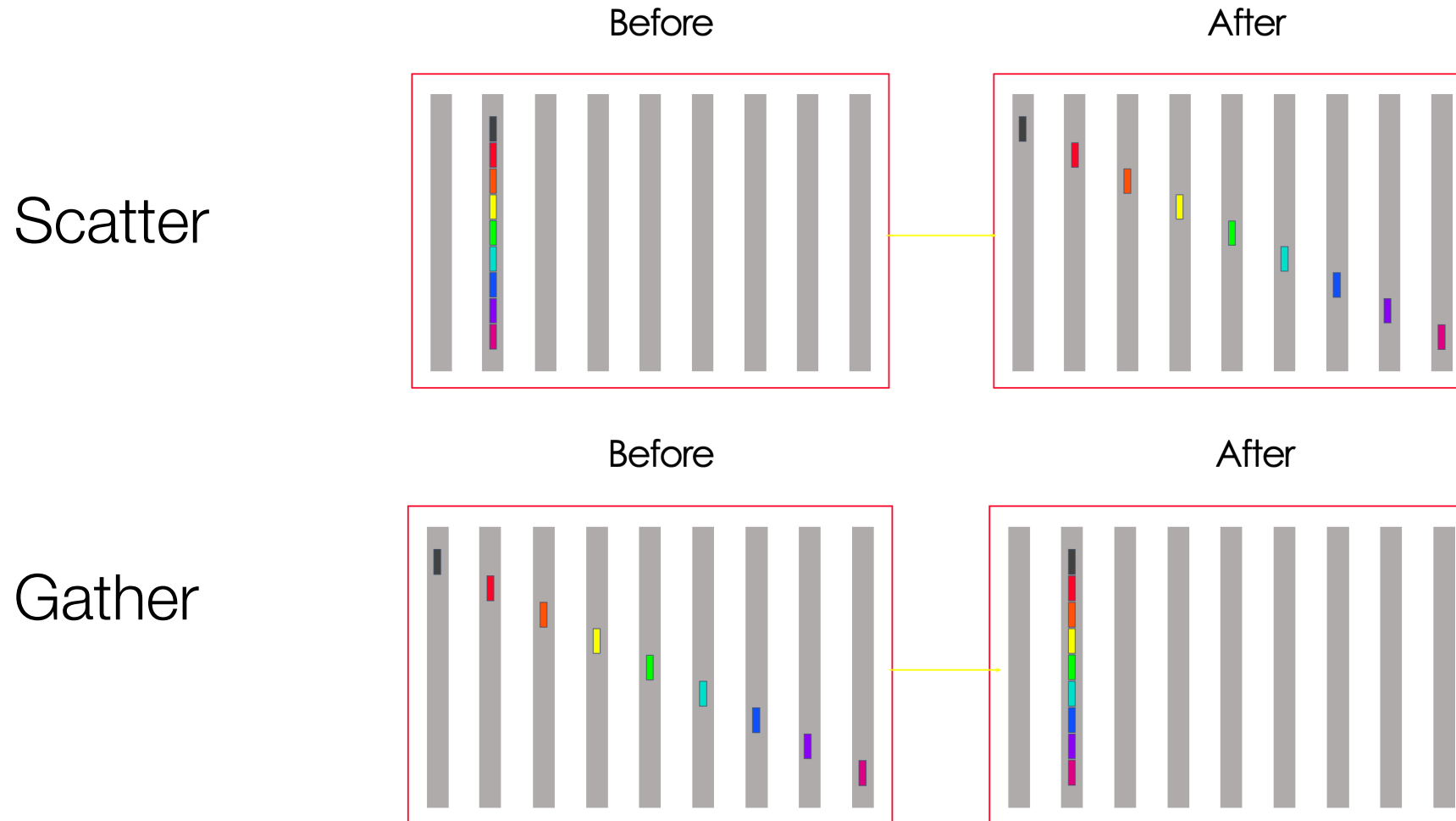
The diagram illustrates the cost of a reduce-scatter operation. It features a central equation with two parts highlighted by colored boxes and arrows. A yellow box highlights the term $(p-1)$, with a yellow arrow pointing to the text "number of steps". A red box highlights the term $\left(\alpha + \frac{n}{p}\beta + \frac{n}{p}\gamma \right)$, with a red arrow pointing to the text "cost per steps". Below these, the expanded equation is shown: $(p-1)\alpha + \frac{p-1}{p}n\beta + \frac{p-1}{p}n\gamma$.

$$(p-1) \left(\alpha + \frac{n}{p}\beta + \frac{n}{p}\gamma \right)$$

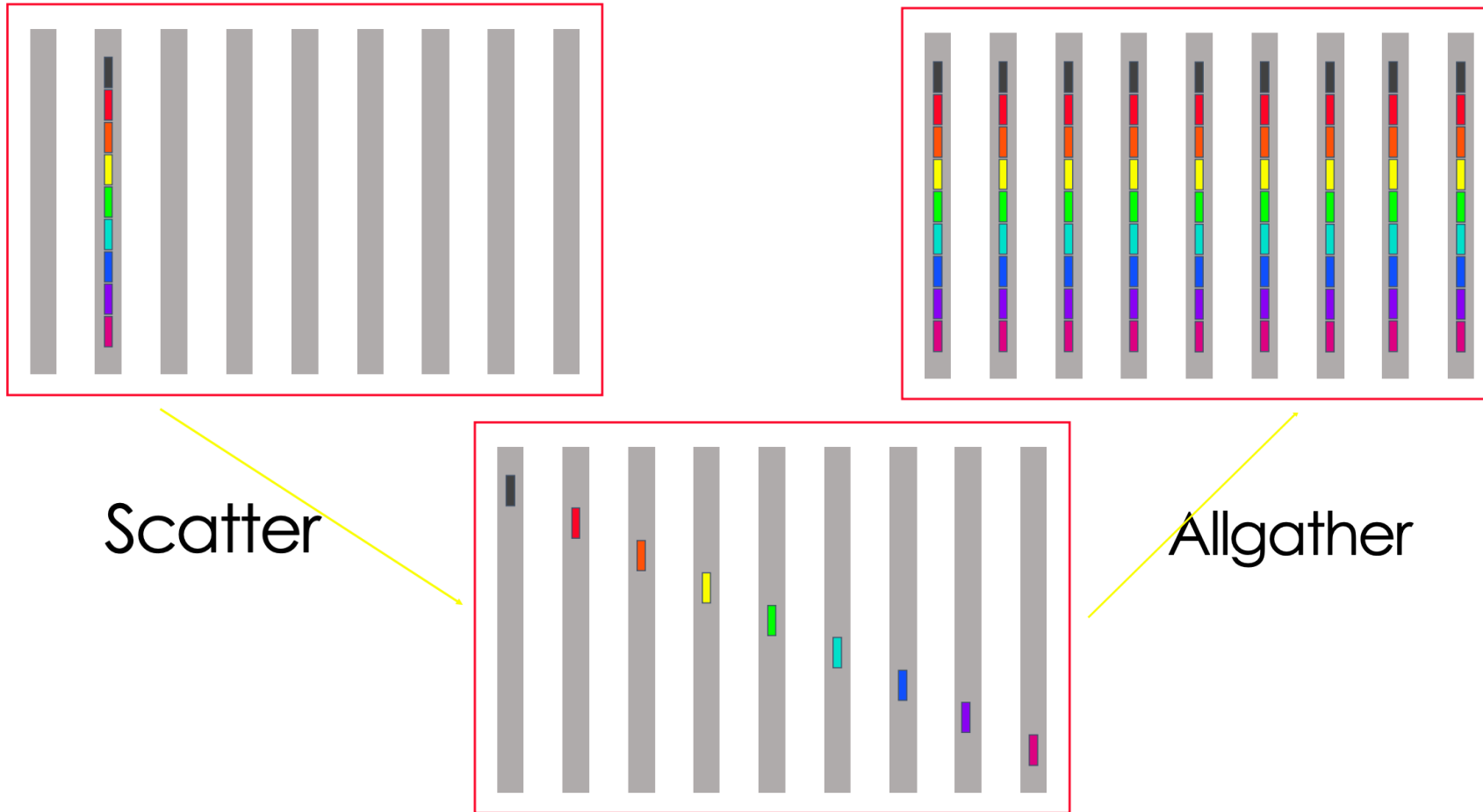
number of steps cost per steps

$$(p-1)\alpha + \frac{p-1}{p}n\beta + \frac{p-1}{p}n\gamma$$

Can ring be better for Scatter and Gather?



Broadcast (large message)



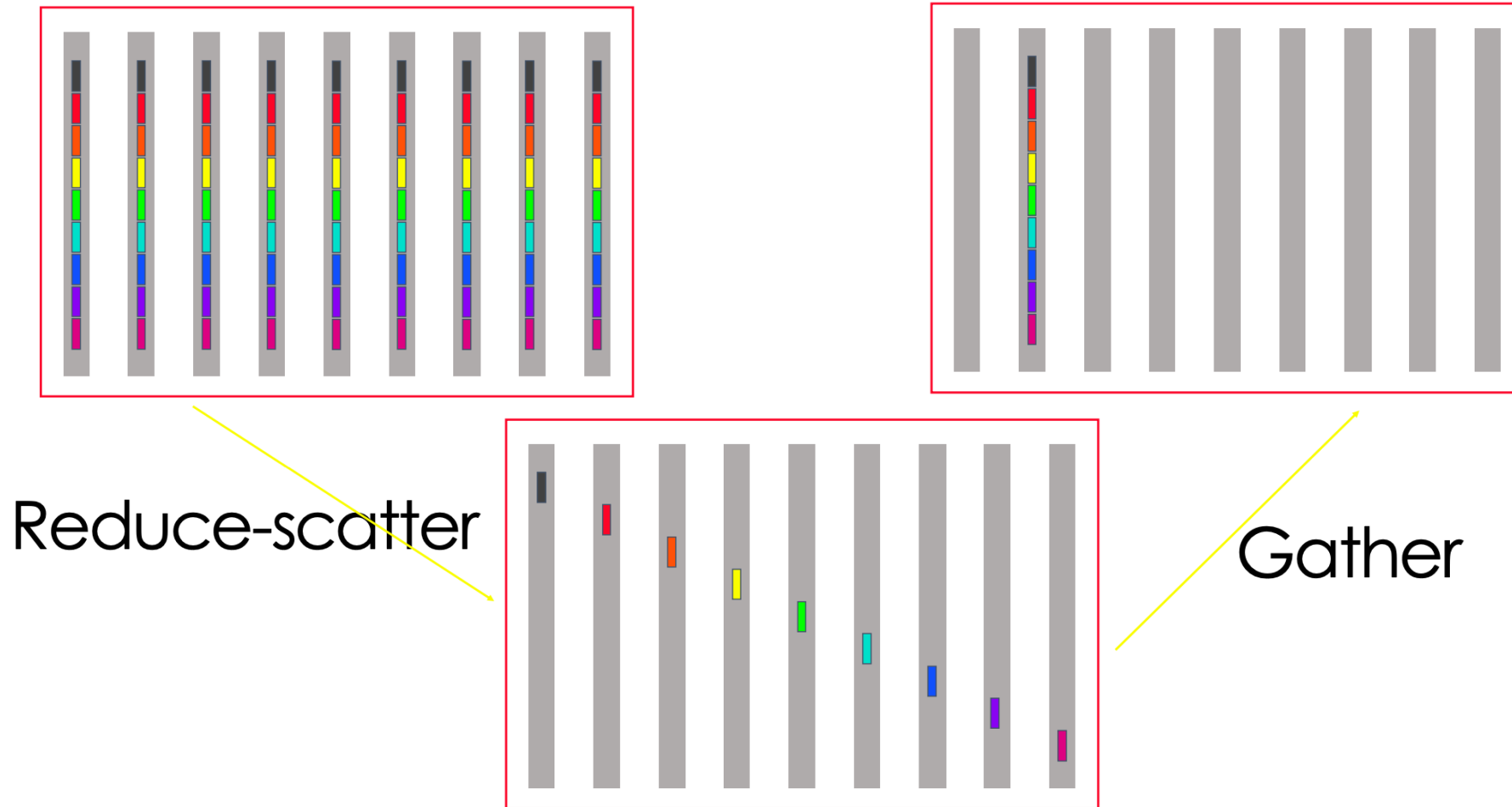
Cost of scatter/allgather broadcast

- Assumption: power of two number of nodes

$$\begin{array}{l}
 \text{scatter} \quad \log(p)\alpha + \frac{p-1}{p}n\beta \\
 \text{allgather} \quad (p-1)\alpha + \frac{p-1}{p}n\beta \\
 \hline
 (\log(p) + p - 1)\alpha + 2\frac{p-1}{p}n\beta
 \end{array}$$

Vs. MST broadcast $\lceil \log(p) \rceil (\alpha + n\beta)$

Reduce (large message)



Cost of reduce-scatter/gather reduce

- Assumption: power of two number of nodes

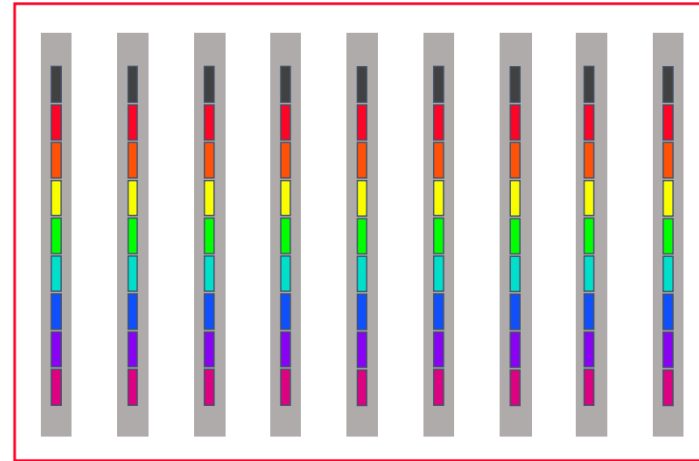
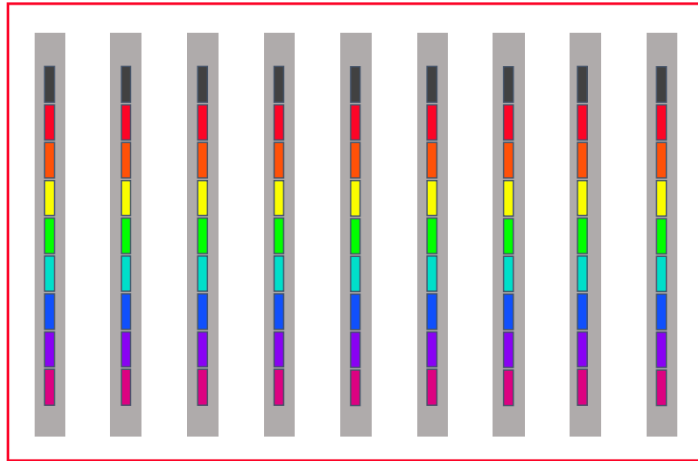
Reduce-scatter $(p-1)\alpha + \frac{p-1}{p}n\beta + \frac{p-1}{p}n\gamma$

gather $\log(p)\alpha + \frac{p-1}{p}n\beta$

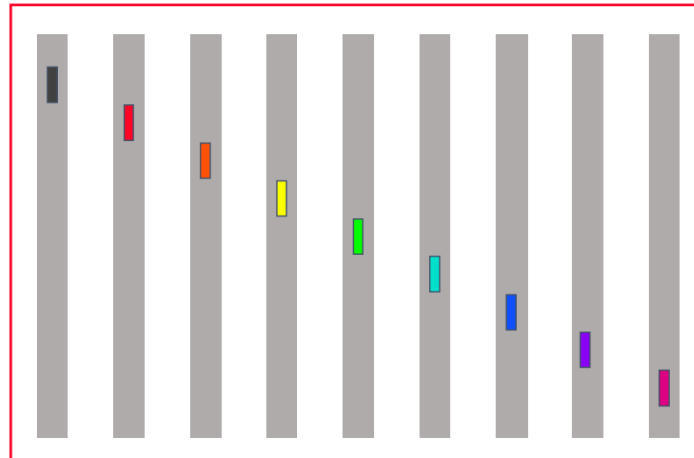
$$(\log(p) + p - 1)\alpha + 2\frac{p-1}{p}n\beta + \frac{p-1}{p}n\gamma$$

Vs. MST reduce $\lceil \log(p) \rceil (\alpha + n\beta + n\gamma)$

Allreduce (large message)



Reduce-scatter



Allgather

Cost of reduce-scatter/Allgather Allreduce

- Assumption: power of two number of nodes

$$\text{Reduce-scatter} \quad (p-1)\alpha + \frac{p-1}{p}n\beta + \frac{p-1}{p}n\gamma$$

$$\text{Allgather} \quad \frac{(p-1)\alpha + \frac{p-1}{p}n\beta}{2(p-1)\alpha + 2\frac{p-1}{p}n\beta + \frac{p-1}{p}n\gamma}$$

$$\text{Vs. Reduce-broadcast allreduce} \quad 2\log(p)\alpha + 2\log(p)n\beta + \log(p)n\gamma$$

Recap

Reduce-scatter

$$(p-1)\alpha + \frac{p-1}{p}n(\beta + \gamma)$$

Scatter

$$\log(p)\alpha + \frac{p-1}{p}n\beta$$

Gather

$$\log(p)\alpha + \frac{p-1}{p}n\beta$$

Allgather

$$(p-1)\alpha + \frac{p-1}{p}n\beta$$

Reduce(-to-one)

$$(p-1+\log(p))\alpha + \frac{p-1}{p}n(2\beta + \gamma)$$

Allreduce

$$2(p-1)\alpha + \frac{p-1}{p}n(2\beta + \gamma)$$

Broadcast

$$(\log(p) + p-1)\alpha + 2\frac{p-1}{p}n\beta$$



Summary of collective communication

- Pros
 - A set of structured / well-defined communication primitives
 - Extremely well-optimized
 - neat, easy to analyze, and easy to understand its performance
- Cons
 - Lack of Fault Tolerance: What if one node (in the ring) is dead?
 - Requires Homogeneity: What if one node computes slower than all other nodes? What if one link has lower bandwidth than the other node?

Collective communication tools/libraries

- MPI: HPC applications and scientific computing
- NCCL: Distributed deep learning across GPUs
- UCX: Unified communication for HPC and deep learning frameworks

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Reference & Credits

- John Wiley & Sons, “Collective communication: theory, practice, and experience”
- Some slides adapted from course slides of DSC204a in UCSD